Electromyographic Correlates of Listening Effort: Differentiating Vestigial Auriculomotor Activity under High Cognitive Load

Master Thesis

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Abstract

This study investigates the physiological recruitment of the human vestigial auriculomotor system, comprising the post-auricular (PAM), superior (SAM), and anterior (AAM) muscles, under conditions of high cognitive load. While these muscles are considered "neural fossils" that once physically oriented the pinnae toward sound sources in ancestral primates, their modern functional role in humans appears repurposed for covert attentional processes [1]. Previous research has established that ear muscle activity reflects the direction of auditory attention in clear listening environments; however, the impact of significant cognitive effort, such as decoding degraded speech, has remained unexplored. Cognitive effort is induced by a Cochlear Implant Simulation.

Utilizing a 2x2x2 repeated-measures design with 13 healthy participants, we manipulated task difficulty and spatial orientation using noise-vocoded cochlear implant (CI) simulations during a dichotic listening task. Electromyographic (EMG) results revealed a statistically significant main effect of task difficulty on PAM activity ($p = .036$), identifying it as a global marker for mental exertion. However, the expected spatial laterality typically seen in clear conditions was absent in the PAM ($p = .981$), suggesting that under high load, the brain prioritizes central decoding effort over peripheral orienting reflexes.

In contrast, the SAM exhibited a more nuanced functional role, characterized by a significant three-way interaction between laterality, difficulty, and position ($p = .045$). This modulation, particularly the strong negative correlation with perceived effort ($\gamma = -0.993$, $p = .007$), suggests an inhibitory "spatial sharpening" strategy where the brain suppresses competing auditory channels to protect the focus of attention. Finally, the AAM provided an essential internal control; while it showed no significant physiological modulation across conditions, its activity correlated nearly perfectly with comprehension levels ($\gamma = 0.993$, $p = .007$), serving as a "success marker" for signal clarity.

Collectively, these findings suggest that under high cognitive load, the auriculomotor system shifts from a primitive spatial tracker to a sophisticated, multi-dimensional biomarker. By triangulating PAM (load), SAM (strategy), and AAM (success), this research provides a physiological framework for tracking listening intent and effort, offering significant implications for the development of "cognitively controlled" hearing instruments.

Zusammenfassung

Diese Studie untersucht die physiologische Rekrutierung des menschlichen vestigialen aurikulomotorischen Systems – bestehend aus dem Musculus auricularis posterior (PAM), superior (SAM) und anterior (AAM) – unter Bedingungen hoher kognitiver Belastung. Während diese Muskeln als „neuronale Fossilien“ gelten, die bei Vorfahren der Primaten die Ohrmuscheln physisch zu Schallquellen ausrichteten, scheint ihre moderne funktionale Rolle beim Menschen für verdeckte (coverte) Aufmerksamkeitsprozesse umgewidmet worden zu sein [1]. Frühere Forschungen haben belegt, dass die Aktivität der Ohrmuskeln die Richtung der auditiven Aufmerksamkeit in klaren Hörumgebungen widerspiegelt; die Auswirkungen erheblicher kognitiver Anstrengung, wie etwa die Dekodierung degradierter Sprache, blieben jedoch bislang ungeklärt. Kognitiver Aufwand wird durch eine Cochlea-Implantat-Simulation induziert.
Unter Verwendung eines 2x2x2-Messwiederholungsdesigns mit 13 gesunden Teilnehmern wurde die Aufgabenschwierigkeit und die räumliche Orientierung mittels Noise-vocoded-Cochlea-Implantat-Simulationen (CI) während einer dichotischen Höraufgabe manipuliert. Elektromyographische (EMG-Ergebnisse) zeigten einen statistisch signifikanten Haupteffekt der Aufgabenschwierigkeit auf die PAM-Aktivität ($p = ,036$), was diesen als globalen Marker für mentale Anstrengung identifiziert. Die erwartete räumliche Lateralität, die typischerweise unter klaren Bedingungen auftritt, fehlte jedoch beim PAM ($p = ,981$). Dies deutet darauf hin, dass das Gehirn unter hoher Belastung der zentralen Dekodierungsleistung Vorrang vor peripheren Orientierungsreflexen einräumt.
Im Gegensatz dazu zeigte der SAM eine differenziertere funktionale Rolle, die durch eine signifikante Dreifach-Interaktion zwischen Lateralität, Schwierigkeit und Position gekennzeichnet war ($p = ,045$). Diese Modulation, insbesondere die starke negative Korrelation mit der wahrgenommenen Anstrengung ($\gamma = -0,993$, $p = ,007$), deutet auf eine inhibitorische Strategie der „räumlichen Schärfung“ hin, bei der das Gehirn konkurrierende Hörkanäle unterdrückt, um den Fokus der Aufmerksamkeit zu schützen. Schließlich bot der AAM eine wesentliche interne Kontrolle; während er keine signifikante physiologische Modulation über die Bedingungen hinweg zeigte, korrelierte seine Aktivität nahezu perfekt mit dem Verständnisniveau ($\gamma = 0,993$, $p = ,007$) und diente somit als „Erfolgsmarker“ für die Signalklarheit.
Zusammengenommen legen diese Ergebnisse nahe, dass sich das aurikulomotorische System unter hoher kognitiver Belastung von einem primitiven räumlichen Tracker zu einem anspruchsvollen, multidimensionalen Biomarker wandelt. Durch die Triangulation von PAM (Belastung), SAM (Strategie) und AAM (Erfolg) bietet diese Forschungsarbeit einen physiologischen Rahmen für die Erfassung von Hörabsicht und -anstrengung, was erhebliche Implikationen für die Entwicklung „kognitiv gesteuerter“ Hörinstrumente hat.

Declaration

I hereby declare that I have authored this work independently, that I have not used sources other than the declared sources and resources, and that I have explicitly marked all material which has been quoted either literally or by content from the used sources. This work has neither been submitted to any audit institution nor been published in its current form.

Saarbrücken, March 3, 2026

Asha Babu

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1 Introduction

1.1 Motivation

Humans and apes do not use ear movements to express emotion, flatten them defensively when scared, or point them toward new and interesting sounds [1]. In contrast, animals like dogs and cats visibly perk and point their ears toward interesting sounds [1]. Unlike many other mammals that swivel their ears toward a sound, humans usually just shift their gaze to look at it. However, recent neuroscience research shows that we still carry an evolutionary leftover, a 'neural fossil.' Even though the movement is too small to see with the naked eye, the muscles around our ears still react and try to orient toward the sounds we pay attention to [2, 1].

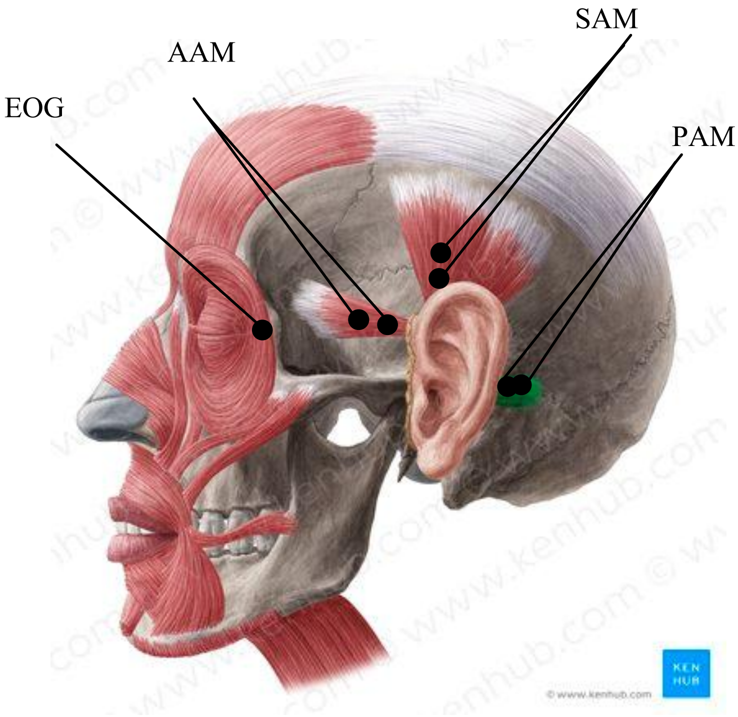


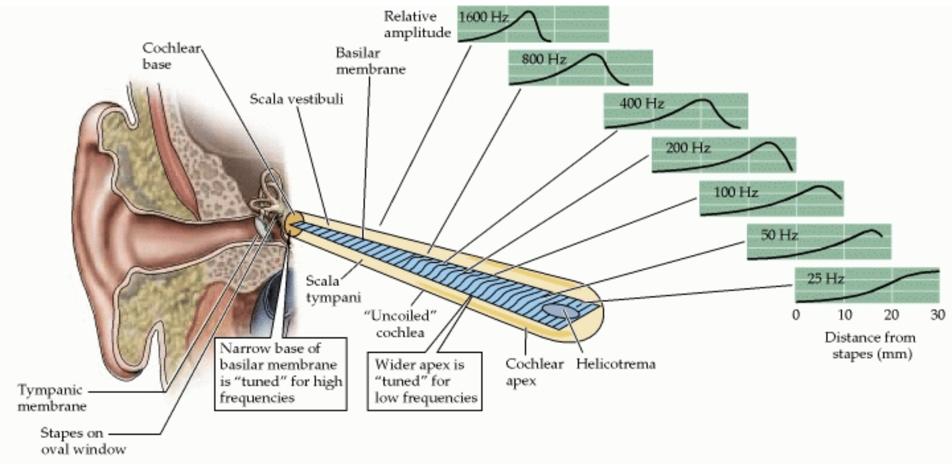
Figure 1.1: Experimental setup for measuring auriculomotor activity. The black dots indicate the placement of bipolar surface electrodes over the three primary extrinsic ear muscles: the anterior (AAM), superior (SAM), and postauricular (PAM) muscles. An additional electrode placement (EOG) is positioned near the eye to track and filter out gaze-shifting artifacts [3].

Using EMG sensors, researchers can measure tiny, involuntary twitches in our ear muscles when we hear a sudden noise. While we cannot consciously swivel our ears toward a sound, the muscle behind our ear still reflexively reacts and points toward the source [4]. The electrical activity of our ear muscles reveals a fascinating mix of automatic reflexes and conscious listening effort, which refers to the intense mental resources and cognitive energy expended to understand speech, particularly in noisy, unclear, or challenging environments [5]. When an abrupt noise catches your attention, it triggers a transient, involuntary reflex often called a "neural fossil" just 70 milliseconds later. This reflexive (exogenous) response is strongly lateralized, meaning the muscles on the ipsilateral side (the ear closest to the sound) react much more forcefully. Figure 1.1 shows an anatomical illustration highlighting the extrinsic muscles around the human ear, specifically the Posterior Auricular Muscle (PAM), Anterior Auricular Muscle (AAM), and Superior Auricular Muscle (SAM). The Post-Auricular Muscle (PAM) serves as a vestigial orienting reflex that remains highly active during sound localisation, particularly when resolving the "cone of confusion", a spatial ambiguity where sounds from the front and back hit both ears with identical timing and volume. While left-right (horizontal) localisation is easily managed by the Superior Olivary Complex through binaural cues such as Interaural Time Differences (ITD) and Level Differences (ILD), front-back (median) localisation requires the brain to analyse spectral cues arising from the physical shape of the pinna (outer ear) [6]. Because the pinna is angled forward, it acts as a frequency filter that muffles or "notches" sounds coming from behind differently than those from the front. The PAM shows particularly strong spikes for rear-originating sounds because it is a primitive evolutionary attempt to physically orient the ear toward the source to better capture these spectral filters. This data is then integrated by the Inferior Colliculus and Auditory Cortex to create a 3D map of the environment, allowing the brain to distinguish whether a sound is a threat from behind or a source from the front. While the PAM is the primary indicator, the anterior (AAM), transverse (TAM), and superior (SAM) auricular muscles also participate; the SAM, for instance, sometimes exhibits a brief suppression in activity on the side of the sound. When you use voluntary attention to actively focus on a specific sound, like straining to hear a voice in a crowded room, these same muscles maintain a higher, sustained level of tension that physically reflects your ongoing listening effort [4].

Building on the foundation laid by Strauss et al. [1], This work relies on their core finding: human auditory attention can be mapped through vestigial ear muscle activity. Using EMG recordings, Strauss et al [1]. showed that muscle activity is strongest in the ear closest to the sound source. Crucially, this ipsilateral response occurred both when participants were involuntarily distracted by unexpected noises and when they actively listened to a story. Their study also established how these muscles react to different spatial locations, specifically comparing front (anterior) and back (posterior) sound placements.

1.2 Evolutionary Context of Auriculomotor Activity

The human cochlea functions as a sophisticated biological frequency analyser, converting mechanical sound waves into neural impulses through spectral decomposition. The cochlea is a fluid-filled, snail-shaped structure located within the temporal bone of the inner ear that functions according to the Place Theory, which states that pitch perception is determined by the specific tonotopic site of maximum vibration along the basilar membrane. Sound enters the fluid-filled, snail-shaped structure, creating travelling waves along the basilar membrane (See Figure 1.2). This membrane is characterised by a strict tonotopic organisation: the narrow, stiff base of the cochlea resonates with high-frequency sounds, while the wider, more flexible apex responds to low-frequency vibrations. As the membrane moves, it displaces the stereocilia of the inner hair cells, which open ion channels to trigger electrical signals in the auditory nerve. This high-resolution mapping allows the healthy ear to distinguish thousands of distinct pitches, providing the "bottom-up" fine-grained detail necessary for effortless



speech perception [7].

Figure 1.2: Tonotopic frequency mapping along the human basilar membrane [8].

In contrast to the thousands of hair cells in a healthy ear, a Cochlear Implant (CI) bypasses damaged sensory cells and directly stimulates the auditory nerve using a limited electrode array, typically with 12 to 22 channels. To simulate this for research, scientists use noise-vocoding, a process that strips away the "fine structure" of speech and leaves only the temporal envelope (the overall volume contour) of specific frequency bands [9]. This processed signal is then used to modulate the noise carriers, yielding a "coarse" representation of the original audio. Because the CI lacks the spectral density of natural hearing, the brain cannot rely solely on the acoustic signal; instead, it must recruit "top-down" cognitive resources, such as working memory, to reconstruct meaning from the degraded temporal cues.

Classic models of attention demonstrate that our cognitive resources are strictly finite; as a task becomes harder, the brain must funnel more energy into central processing. Because of this limit, it is currently unknown whether the brain continues to power vestigial reflexes, such as spatial ear orienting, when cognitive demands peak [10, 11]. If an auditory task is difficult enough, the central nervous system might actually suppress these tiny ear muscle contractions, essentially prioritising conscious listening effort over an ancient motor reflex [12, 13]. To investigate how task difficulty affects spatial hearing reflexes, we designed a three-way repeated-measures experiment. We looked at three variables: Laterality (ipsilateral vs contralateral), Position (front vs back), and Task Difficulty (easy vs difficult). Crucially, we manipulated task difficulty by altering the number of channels and the frequency range in a cochlear implant Simulation application to produce cochlear implant (CI)- simulated audio (Difficult and Easy) by the University of Granada [9]. By applying a cognitive load element to the spatial framework established by Strauss et al [1]. This study aims to determine if the vestigial orienting response remains robust during difficult tasks or if the brain's need for cognitive resources shuts the reflex down entirely.

2 Problem Analysis and Goals

2.1 State of the Art

The current state of the art in tracking selective auditory attention and listening effort relies on a combination of central and peripheral physiological biomarkers. Historically, central measures such as electroencephalography (EEG) have dominated the field. Neural tracking techniques, such as stimulus reconstruction from cortical activity, have proven highly effective in decoding which speaker a listener is attending to in a "cocktail party" scenario [14]. Similarly, pupillometry has become a gold-standard autonomic biomarker for quantifying "listening effort," with pupil dilation reliably reflecting the cognitive load required to understand degraded speech [15].

However, these traditional measures possess distinct limitations: EEG requires extensive, obtrusive setups and is highly susceptible to motion artifacts, while pupillometry is heavily confounded by ambient lighting changes. In response, recent research has shifted toward peripheral motor systems as robust, non-invasive alternatives. Building on the foundational discovery that vestigial auriculomotor activity reflects the lateral focus of attention [1], the surface electromyography (EMG) of the postauricular muscle (PAM) has emerged as a critical biometric. Under normal hearing conditions, the state of the art demonstrates that PAM activity strongly and continuously correlates with the spatial direction of attention, firing more robustly on the side of the attended stimulus during both exogenous and endogenous orienting tasks.

In parallel with developments in attentional tracking, the audiological study of listening effort has established standardised methodologies for simulating hearing impairment. To investigate cognitive load without the confounding clinical variability inherent in actual hearing-impaired populations, researchers frequently utilise noise-band vocoders [16]. These acoustic models artificially replicate the severe spectral blurring, limited channel resolution, and degraded temporal cues characteristic of Cochlear Implant (CI) transmission. By presenting this CI-simulated audio to normal-hearing listeners, the current state of the art allows researchers to reliably induce, isolate, and manipulate high cognitive load in strictly controlled experimental environments [17].

2.2 Recent Advances in Research

Lately, research on the vestigial auriculomotor system has shifted away from using simple, sudden noises such as clicks and chirps toward more realistic, continuous speech tests. A breakthrough in this area showed that the postauricular muscle (PAM) does not just twitch in response to sudden sounds. Instead, it actively stays tense when we voluntarily focus on someone talking for an extended period [1]. Researchers have successfully tracked this sustained muscle activity during classic 'cocktail party' scenarios, where a listener has to tune into one voice among competing speakers [14]. These recent studies confirm that our peripheral ear muscles act as a continuous, physical readout of our spatial attention, firing much more strongly on the side of the voice we are trying to hear.

At the same time, cognitive hearing science has started to focus heavily on the idea of 'listening effort' [18]. Speech perception is increasingly recognised as an active cognitive process rather than a passive auditory one. When the acoustic signal is compromised, whether by environmental noise or sensorineural hearing loss, listeners must compensate by recruiting 'top-down' cognitive resources, such as working memory, to reconstruct meaning from fragmented input. To study this in the lab, researchers often use noise-band vocoders to mimic the distorted sound of a cochlear implant (CI). This setup reliably forces a high cognitive load, letting scientists observe how the brain shifts its limited resources when simply trying to hear becomes an exceptionally difficult task [9].

Currently, models of PAM activation are based on clear, normal-hearing conditions, leaving a significant gap in how these reflexes behave under heavy cognitive load. Because the brain has a finite pool of processing resources [11], demanding tasks like decoding degraded, CI-simulated speech force the central nervous system to prioritise conscious listening effort over secondary processes, refer to downstream cognitive tasks that the brain would normally perform if it weren't so exhausted by the act of listening. It is fundamentally unknown whether vestigial ear muscle contractions persist during this resource depletion. If the brain suppresses these spatial motor reflexes to conserve energy for cognitive decoding, then the PAM's reliability as an attentional biomarker is directly bound by task difficulty.

The overarching objective of this study is to determine how varying levels of cognitive load impact vestigial spatial orienting. We utilised a 2x2x2 repeated measures design focusing on Laterality (ipsilateral vs contralateral), Position (front vs back), and Task Difficulty (normal audio vs CI-simulated audio).

Specifically, we aim to:

- This study evaluates whether the Posterior (PAM), Superior (SAM), and Anterior (AAM) Auditory Maps maintain their characteristic hemispheric asymmetry during the processing of CI-simulated speech, or if the increased cognitive load associated with spectral degradation triggers a shift toward bilateral recruitment.
- Evaluate the potential interaction between the spatial location of competing auditory streams and the cognitive effort required to process them.

3 Materials and Methods

3.1 Participants

The study included 13 healthy participants (5 women, 8 men; average age 25.62 ±1.6 years). All participants provided written informed consent before the sessions began. The research was conducted in full accordance with the Declaration of Helsinki [19].

3.2 Electrophysiological Setup

Following the methodology described by Strauss et al. [1], the skin behind both ears was cleaned with alcohol and gently abraded to reduce skin impedance. We recorded surface EMGs using non-recessed Ag/AgCl electrodes from BioMed Electrodes (USA). As shown in Figure 1.1. All recording sites were fitted with larger 6 mm electrodes (BME6). The signals were digitised at a sampling rate of 4800 Hz with a 24-bit resolution per channel using a g.USBamp system (g.tec GmbH, Austria). Alongside the EMG data, we continuously monitored eye movements (EOG). All subsequent signal processing was carried out using MATLAB 2023a (MathWorks, USA).

3.3 Auditory Stimuli

To ensure a high degree of ecological validity and continuous listener engagement, podcasts were chosen as the primary auditory stimuli. The selection criteria required high-quality, single-speaker narratives without background music or extreme acoustic modulations to prevent unwanted electrophysiological artefacts. We selected the *Leonardo English* podcast [20], which provides clear, educational storytelling across a wide variety of distinct podcasts (e.g., science, food). Of the 23 collected episodes, 18 were selected and edited into cohesive 5-minute narrative segments. Having a diverse range of topics was crucial, as it allowed us to present two entirely distinct, competing streams of information simultaneously during the continuous-attention paradigms.

3.4 Cochlear Implant (CI) Simulation

A core methodological objective of this study was to reliably induce a "difficult" listening condition by processing the audio through a Cochlear Implant (CI) simulation. All major CI systems share the same fundamental architecture: bypassing damaged hair cells in the inner ear to provide sound perception via direct electrical stimulation of the auditory nerve [21]. However, they differ significantly in their coding strategies. We analysed the audio processing mechanisms of the three leading manufacturers:

- **Cochlear Limited**: Utilises the Advanced Combination Encoder (ACE), a "peak-picking" strategy that analyses 22 frequency bands but only stimulates the 8–12 highest amplitude maxima at any given moment. This reduces electronic interference and optimises speech clarity.
- **MED-EL**: Employs Fine Structure Processing (FSP/FS4) using rate coding at the cochlear apex to mimic the natural timing of incoming sound waves, primarily enhancing pitch perception and music richness.
- **Advanced Bionics** (AB): Focuses on spectral resolution using "current steering" (HiRes Fidelity 120), which varies power between physical electrodes to create virtual channels and high-definition sound.

We designed our difficult condition to mimic the specific peak-picking processing patterns of these cochlear implants.

Finding a reliable vocoder to achieve this specific degradation proved challenging. Standard vocoders manipulate the number of frequency channels to mimic the cochlear mapping, low frequencies to the apex and high frequencies to the base in the basilar membrane. However, custom MATLAB-designed vocoders generated artifacts that rendered the audio unrealistically distorted rather than reliably degraded. Furthermore, the CI simulation model available in MATLAB Simulink [22], as shown in Figure 3.4.3, lacked user-friendly parameter adjustments, and the JUCE audio framework proved overly complex and inaccessible for our specific manipulation needs.

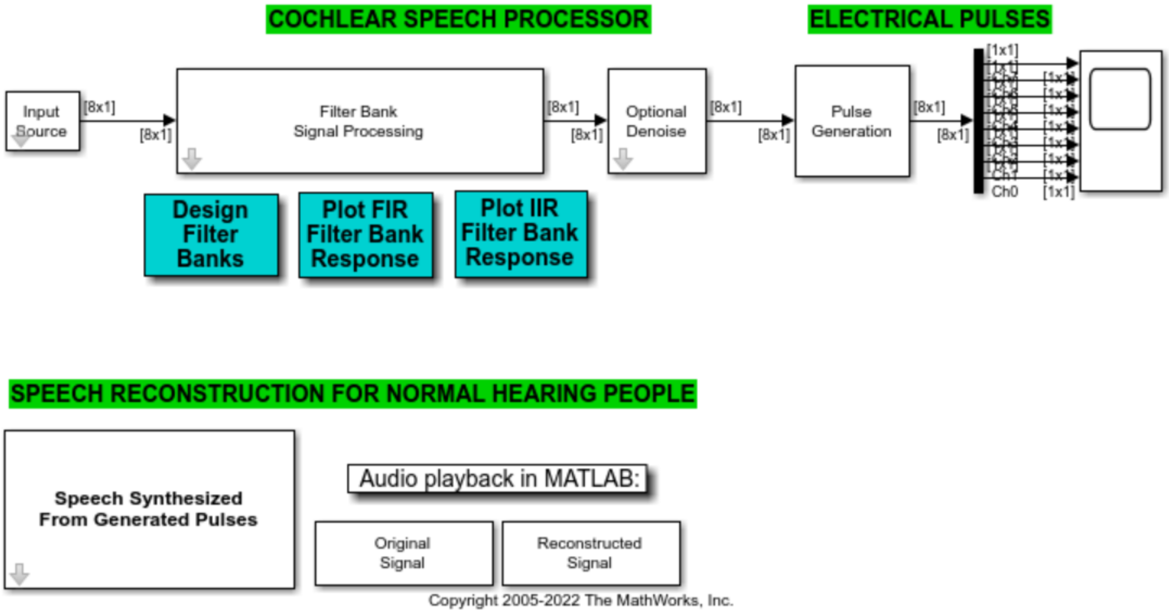


Figure 3.4.1: Simulink model of Cochlear speech processor [22]

Ultimately, the audio was processed using the *Cochlear Implant Simulation (Version 2.0)* developed by the University of Granada (2004) [23]. The standalone application shown in Figure 3.4.2 was selected because it provided a highly accessible interface that enabled precise, manual adjustment of all critical parameters (such as the number of channels and frequency ranges). This flexibility was essential for accurately mimicking the Cochlear Simulation strategy and successfully standardising the cognitive load for our difficult listening condition.

Figure 3.4.2 Cochlear Implant Simulation Software [23]

The selected software serves as a versatile platform that enables normal-hearing individuals to perceive audio as it would sound similarly through a physical cochlear implant. Designed to run as a lightweight, standalone executable (ci_sim.exe) on standard Windows operating systems, the program bypasses complex installation procedures, see Figure 3.4.2. Its core workflow consists of four streamlined stages: preparation of the input audio (via loaded WAV files or live microphone recording), configuration of the specific implant parameters, simulation of the processed signal, and finally, storage of the synthesised audio for experimental playback.

Under the hood, the software operates on a highly detailed two-stage Analysis-Synthesis model. The Analysis Block essentially emulates the external processor of a real implant. It runs the incoming audio through a filter bank, performs envelope detection (using IIR filters with rectification or FIR filters with Hilbert transforms), adapts the dynamic range, and generates electrical stimulation pulses while accounting for the interaction between the electrodes and the neural endings. Following this, the Synthesis Block translates this simulated neural activity pattern back into a perceivable acoustic signal. It does this by filtering an excitation signal, multiplying it by the corresponding envelope for each channel, and summing the results to form the final degraded output.

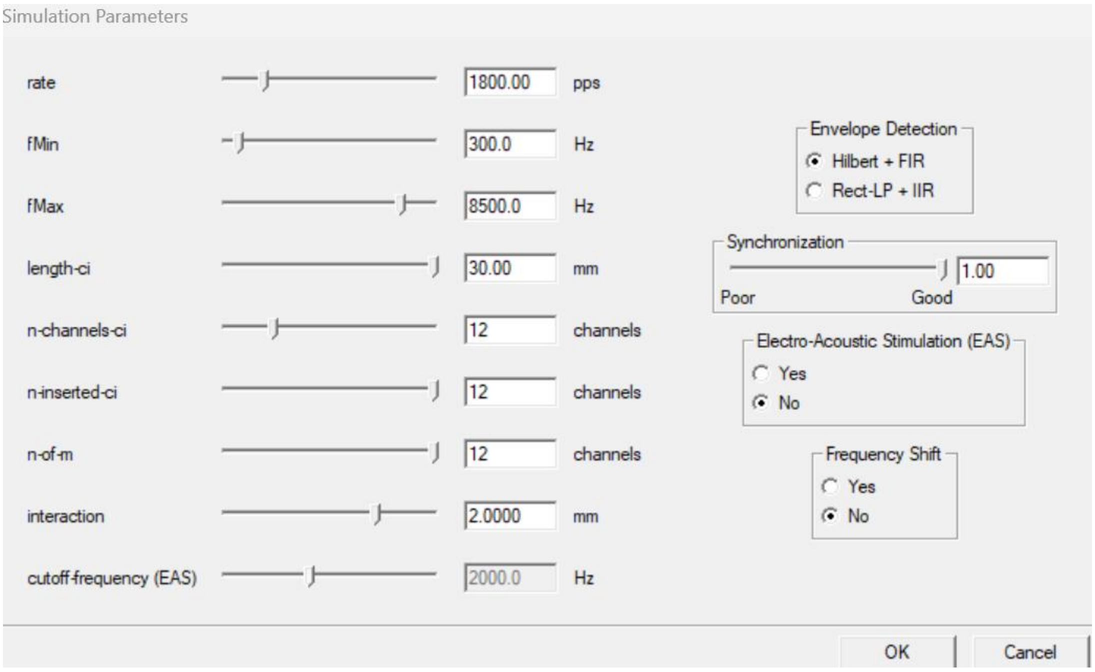


Figure 3.4.3 Cochlear Implant Simulation Parameters [23]

What made this software invaluable for our study was the depth of its customizable parameters in Figure 3.4.3, which allowed us to precisely mimic target CI devices. For instance, the tool allows researchers to manipulate the spectral frequency range (fMin and fMax) and the physical geometry of the implant (such as length and the number of inserted electrodes). Because pitch perception relies heavily on the cochlea's place theory, adjusting this geometry directly influences the distance between electrodes and the resulting frequency shifts. Crucially, the software supports an *n-of-m* coding strategy, which only activates the *n* channels with the highest energy out of *m* available channels during each cycle.

3.5 System Validation and Limitations

The developers originally validated the accuracy of this simulation tool by testing it on a cohort of seven patients with MED-EL cochlear implants. In these trials, patients compared original audio sentences to synthesised versions processed with varying parameters. The results demonstrated a clear "knee effect" in the data: patients perceived the synthesised simulation as just as clear as the original audio right up until the simulation parameters fell below the actual technological capabilities of their physical implants. This confirmed that the software accurately mimics real-world device degradation.

However, when interpreting the stimuli used in our study, it is important to acknowledge the inherent limitations of using a computerised simulation. The software relies on an idealised physiological model. It assumes a perfectly programmed processor map, a structurally normal cochlea, and flawless surgical electrode insertion—failing to account for real-world complications like ossified cochlea’s or variations in a patient's intensity resolution due to surviving neuron counts. Furthermore, the model strictly simulates peripheral auditory processing; it cannot account for signal processing that occurs higher up in the auditory brainstem or at the cortical level, nor does it factor in the profound impact of a real patient's long-term auditory learning and neuroplasticity.

3.6 Experimental Design

To investigate the interaction between cognitive load and spatial attention, the study utilised a 2x2x2 within-subjects repeated measures design. The experiment evaluated three independent variables: Laterality - whether the attended stream was ipsilateral or contralateral to the recorded PAM, Position - whether the competing sound sources were located in the front or back of the spatial fields, and Task Difficulty is determined by the ease or difficulty of the conditions. To establish the different levels of cognitive load, we processed the audio through the CI simulation software under two distinct parameter settings. The easy condition was designed to provide higher auditory resolution by using 16 electrodes across a wide frequency range spanning from 250 Hz to 8 kHz (Figure 3.6.1). To induce the difficult condition, we significantly degraded the signal by limiting the simulation to 6 electrodes and compressing the frequency range to a narrow band between 500 Hz and 6 kHz (Figure 3.6.2)

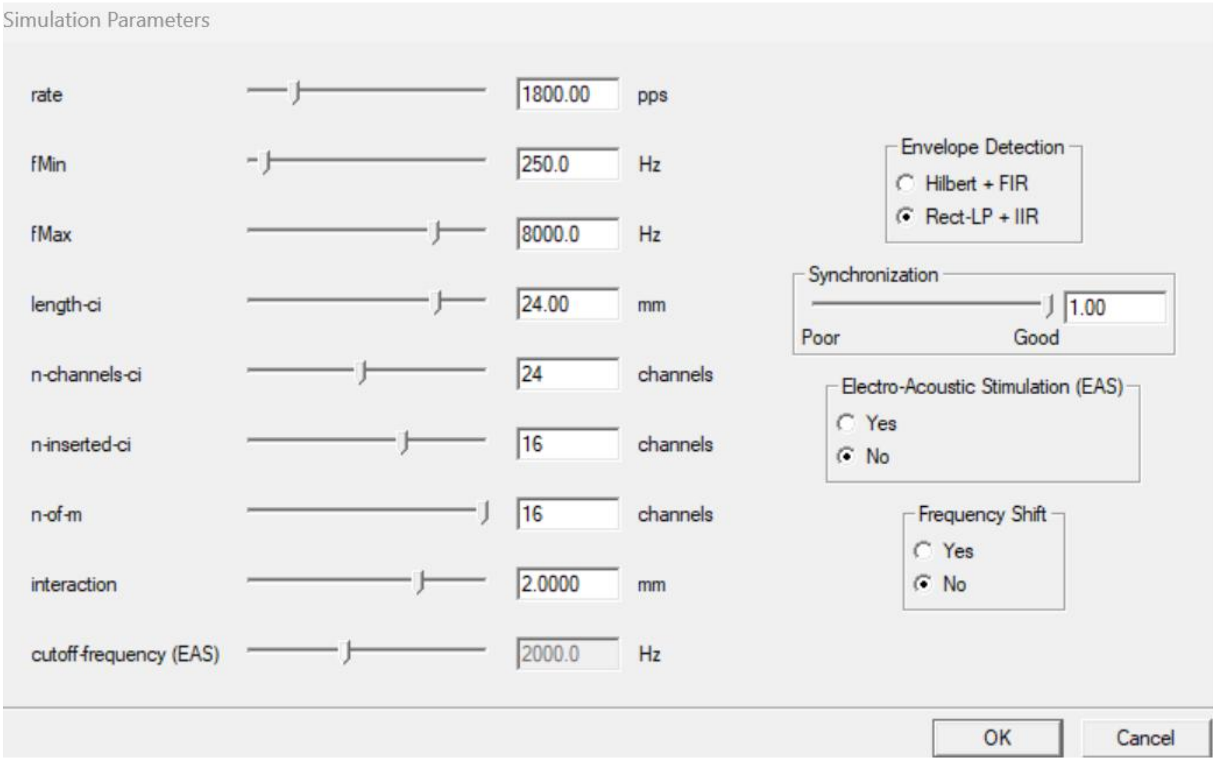


Figure 3.6.1 Easy condition Parameters **[23]**

To finalize the signal processing parameters, envelope detection was maintained using an Infinite Impulse Response (IIR) filter. To accurately model a healthy auditory nerve and ensure proper neural firing patterns, the synchronization parameter was set to 'good.' Additionally, we strictly disabled the electroacoustic simulation feature. While electroacoustic models combine electrical stimulation with residual natural hearing, enabling it in this context would have produced unrealistically clear speech, undermining our goal of creating a rigorous, purely electrical cochlear implant simulation. Finally, artificial frequency shifting was not enabled, ensuring that the spectral mapping remained entirely faithful to the baseline electrode geometry [23].

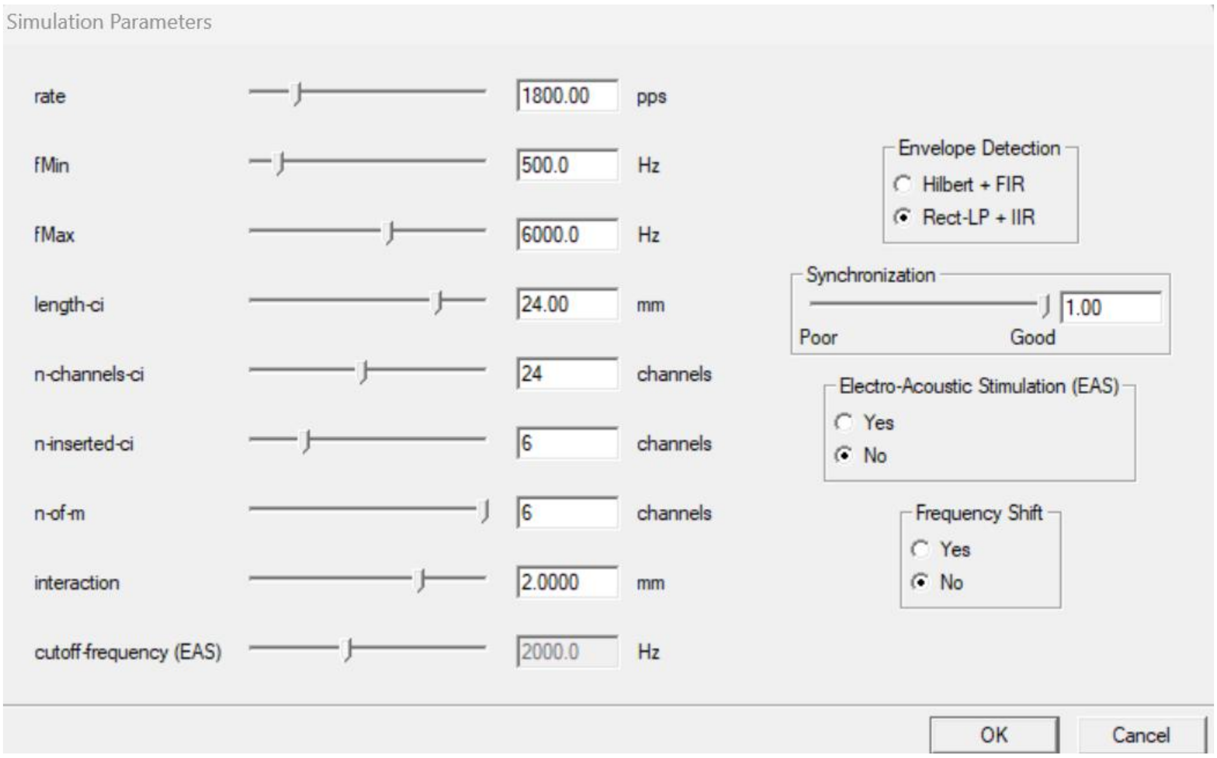


Figure 3.6.2 Difficult condition Parameters **[23]**

3.7Experimental Setup and Procedure

Auditory streams were delivered via a KH120A (Neumann, Germany), a four-speaker array positioned at ±30 ° and ±120 °. Participants were seated in the centre of the sound-attenuated room. The audio streams were presented through four loudspeakers, all placed 1 meter from the participant's head, as illustrated in Figure 3.7.1. Throughout the experiment, participants were instructed to maintain their gaze on a central fixation ball as shown in Figure 3.7.2. This was a critical control measure to prevent the oculo-auricular phenomenon, which is an involuntary vestigial reflex where the muscles of the outer ear (the pinnae) contract in response to lateral eye movements, essentially "pointing" the ears in the direction of visual gaze **[24]**, ensuring that any recorded ear muscle activity was not simply an artefact of eye movements. Horizontal eye movements were calibrated to ensure that gaze-related artefacts could be accurately identified and partitioned from the neural data. This was achieved by having participants toggle their gaze between the centre and peripheral speakers ±30° for five trials. The resulting data (see Figure 3.8.2) was used to define the scale factor for gaze position relative to the speaker array, effectively 'grounding' the EOG signal in physical degrees.

During each block, participants were directed to actively listen to a target story from one specific spatial location while simultaneously ignoring a competing story presented from the opposite side. The experimental paradigms were controlled via MATLAB. Participants listened to 5-minute stories presented at 50 dBA, which was normalised using an audio meter. This was immediately followed by comprehension questions to verify active listening. The speaker output was managed via Studio One audio software. To create the competing-speaker paradigm, each audio file comprised two concurrent podcast stories and an embedded trigger channel. We utilized a total of nine distinct audio tracks throughout the session, which included one initial training block and eight experimental blocks. The head of the participants was fixed with a fixation rod located above the seat, see Figure 3.7.1

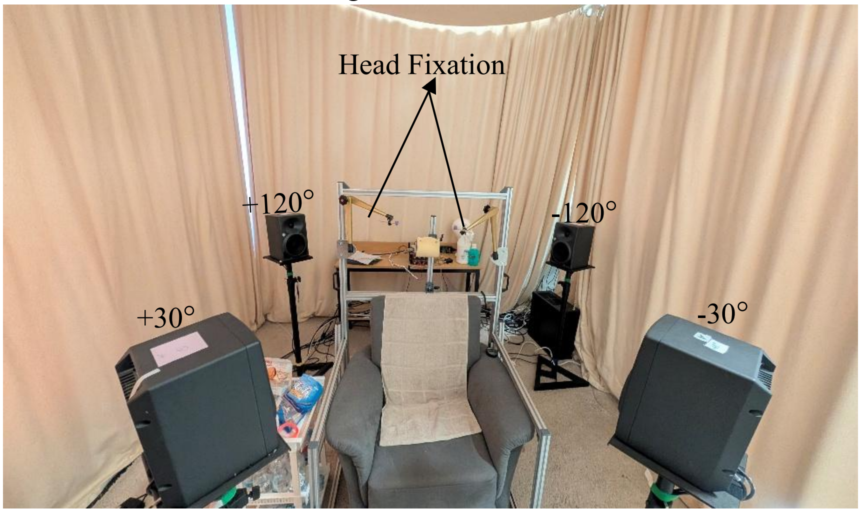


Figure 3.7.1 Experiment setup with speakers


Figure 3.7.2 Gaze fixation ball to reduce eye movements

Channel	Electrode
1	Time
2	PAM R+
3	PAM R-
4	PAM L+
5	PAM L-
6	SAM R+
7	SAM R-
8	SAM L+
9	SAM L-
10	AAM R+
11	AAM R-
12	AAM L+
13	AAM L-
14	EOG R
15	EOG L
16	Not Used
17	Not Used
18	Trigger

Table 3.7.1 Channel Map

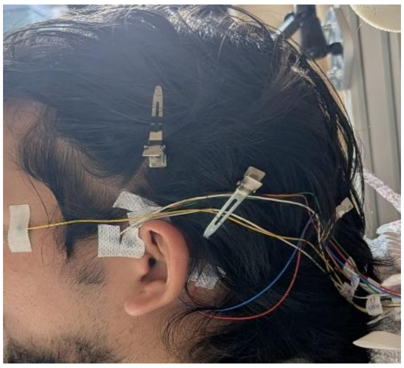


Figure 3.7.3 Participant after with attached the electrodes

Given above is Figure 3.7.3 of a participant after attaching the EOG L, PAM L+, PAM L-, SAM L+, SAM L-, AAM L+, and AAM L-. After the participant was seated comfortably, electrodes were connected based on the channel map in Table 3.7.1. Loading a topographic montage was unnecessary because EEG measurements were absent. Impedance levels were kept below 5 kΩ to ensure signal integrity.

We also play a training session to familiarise them with the audio. To ensure high participant engagement, we allowed subjects to choose between two different podcasts and choose their preferred story. They were then informed which speaker would be broadcasting their chosen narrative. Following the protocol of our previous study, participants were instructed to maintain focus on a fixated ball while keeping their head stabilised with a fixating rod. Upon completion of each trial, participants provided subjective ratings of their perceived listening effort, task exhaustion, and overall speech intelligibility. To ensure sustained auditory attention and validate the processing of the podcast content, a five-item multiple-choice questionnaire (MCQ) was administered as an objective measure of comprehension. The process was repeated with a new story, and the target listening direction was switched to another speaker location or side. For our analysis, muscle activity was quantified by breaking the data into consecutive 10-second segments to track temporal changes.

3.8 Data Processing and Statistical Analysis

Offline data processing and signal analysis were performed using MATLAB b2023. To isolate the relevant muscle activity, auricular EMG signals were filtered between 10 and 1000 Hz using a 2000th-order finite impulse response (FIR) filter, along with a 2nd-order infinite impulse response (IIR) notch filter at 50 Hz to eliminate ambient power-line noise. Simultaneously, horizontal EOG signals were filtered from 0.01 to 20 Hz using a 2nd-order IIR filter to track eye movements. To ensure signal integrity, a bandpass filter with a 10–1,000 Hz bandwidth was applied to the raw data. This frequency range was specifically selected to isolate the high-frequency components of the muscular response while intrinsically attenuating low-frequency artefacts (< 10 Hz), such as those generated by eye blinks, ocular drifts, and gross head movements. Following this spectral filtering, the signal energy for each condition was calculated and normalised to each participant's Maximum Voluntary Contraction (MVC) to account for inter-subject anatomical variability and electrode impedance differences.

To prevent any phase distortion, all filtering operations were executed using MATLAB's filtfilt function for zero-phase filtering. The filtered signals were subsequently downsampled to 2400 Hz. Raw signal and processed signal are given in Figure 3.9.1. All statistical analyses were conducted using JASP software on a Windows operating system. A repeated measures Analysis of Variance (ANOVA) was performed on the normalised PAM EMG activity to assess the main effects and interactions of the within-subjects factors: stimulus-muscle correspondence (ipsilateral vs contralateral responses), anteriority (front vs back speakers), and task difficulty. The significance level was established at $\alpha = 0.05$.

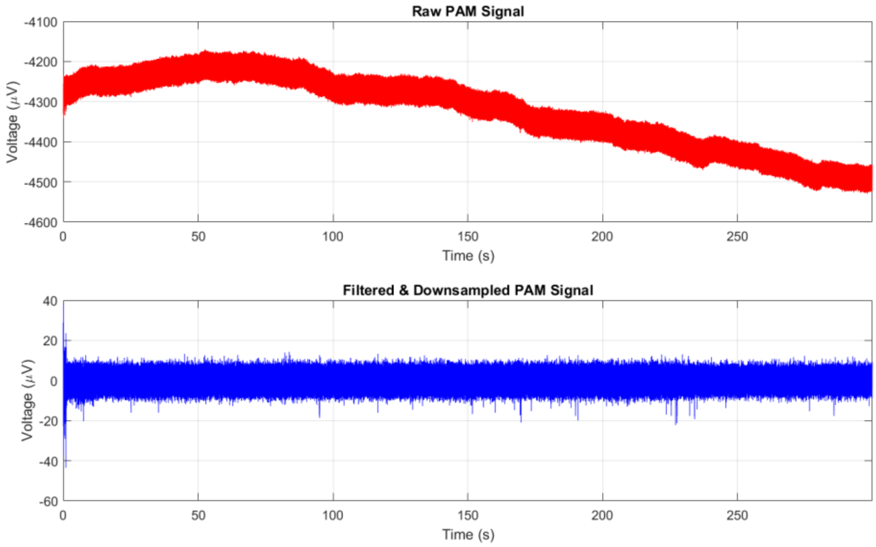


Figure 3.8.1 Filtered data.

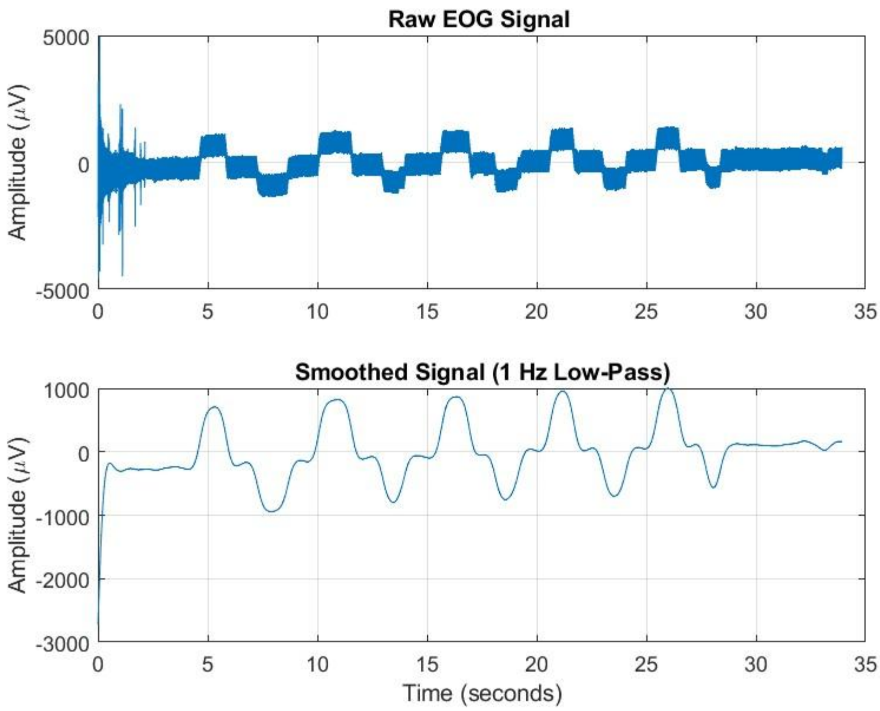


Figure 3.8.2. EOG Calibration signal processed.

4 Results

4.1 Vestigial Auriculomotor (PAM) Response

To investigate the role of voluntary, goal-directed attention, we implemented a classic dichotic-listening paradigm (based on the work of Hillyard et al.) [25]. In this setup, two competing short stories were played simultaneously, localised to either the two front or the two back speakers.

To investigate the nuanced relationship between spatial location and cognitive load on vestigial auriculomotor activity, a three-way repeated measures ANOVA ($p < 0.05$) was performed on the normalised electromyography (EMG) data of the postauricular (PAM), superior (SAM), and anterior (AAM) auricular muscles. This analysis utilised Laterality (ipsilateral vs contralateral), Position (front vs back), and Task Difficulty (easy vs difficult) as within-subjects factors to isolate the specific drivers of ear-muscle activation.

Focusing specifically on the PAM data, the results revealed a statistically significant main effect of Difficulty ($F(1, 12) = 5.549$, $p = .036$, $\eta^2 = .006$), suggesting that the intensity of cognitive demands acts as a primary modulator of PAM activity, independent of where the stimulus is located or which side is being measured.

In contrast, no significant main effects were observed for Anteriority ($F(1, 12) = 0.959$, $p = .347$, $\eta^2 = .018$) or Laterality ($F(1, 12) = 0.001$, $p = .981$, $\eta^2 < .001$), indicating that these spatial factors did not independently drive muscle response. Furthermore, the analysis found no significant two-way interactions specifically between Laterality and Difficulty ($p = .542$), Laterality and Anteriority ($p = .572$), or Difficulty and Anteriority ($p = .339$) and the three-way interaction also failed to reach statistical significance ($F(1, 12) = 3.162$, $p = .101$, $\eta^2 = .061$). Collectively, these findings highlight that while cognitive load reliably triggers vestigial PAM activation, this response remains remarkably uniform across different spatial and lateral orientations.

Within-Subjects Effects (significant at $p<0.05$)						
Cases	Sum of Squares	df	Mean Square	F	p	η^2
Laterality	4.002×10^{-6}	1	4.002×10^{-6}	5.689×10^{-4}	.981	7.881×10^{-7}
Residuals	0.084	12	0.007			
difficulty	0.028	1	0.028	5.549	.036	0.006
Residuals	0.061	12	0.005			
Anteriority	0.091	1	0.091	0.959	.347	0.018
Residuals	1.139	12	0.095			
Laterality ✻ difficulty	0.007	1	0.007	0.393	.542	0.001
Residuals	0.218	12	0.018			
Laterality ✻ Anteriority	0.039	1	0.039	0.337	.572	0.008
Residuals	1.405	12	0.117			
difficulty ✻ Anteriority	0.039	1	0.039	0.993	.339	0.008
Residuals	0.470	12	0.039			
Laterality ✻ difficulty ✻ Anteriority	0.312	1	0.312	3.162	.101	0.061
Residuals	1.183	12	0.099			

Note. Type III Sum of Squares

Table 4.1.1 ANOVA Table of PAM

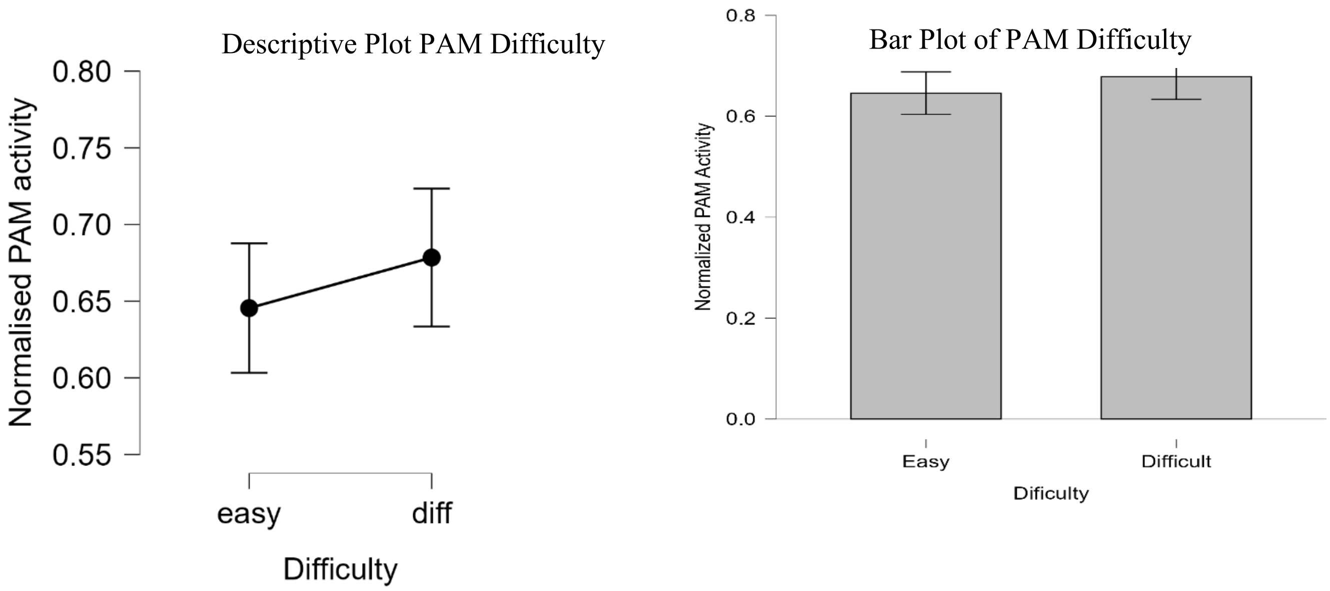


Figure 4.1.1 Normalised PAM Activity as a Function of Task Difficulty.

Figure 4.1.1 illustrates the primary influence of cognitive load on vestigial auriculomotor activity, specifically highlighting the normalised postauricular muscle (PAM) response across varying levels of task difficulty. A clear upward trend is observed as task demands transition from the "Easy" to the "Difficult" condition, with mean normalised activity rising from approximately 0.64 to 0.68, respectively. This elevation in the "Difficult" condition reinforces the ANOVA's statistically significant main effect of Difficulty ($p = .036$), suggesting that increased cognitive engagement reliably recruits the PAM reflex. While the overlap in error bars suggests some inter-individual variability in the magnitude of the response, the consistent positive slope confirms that more demanding tasks elicit higher levels of muscle activation. Consequently, these data provide evidence that the PAM reflex serves as a sensitive, albeit subtle, physiological marker for cognitive effort, independent of spatial or lateral factors.

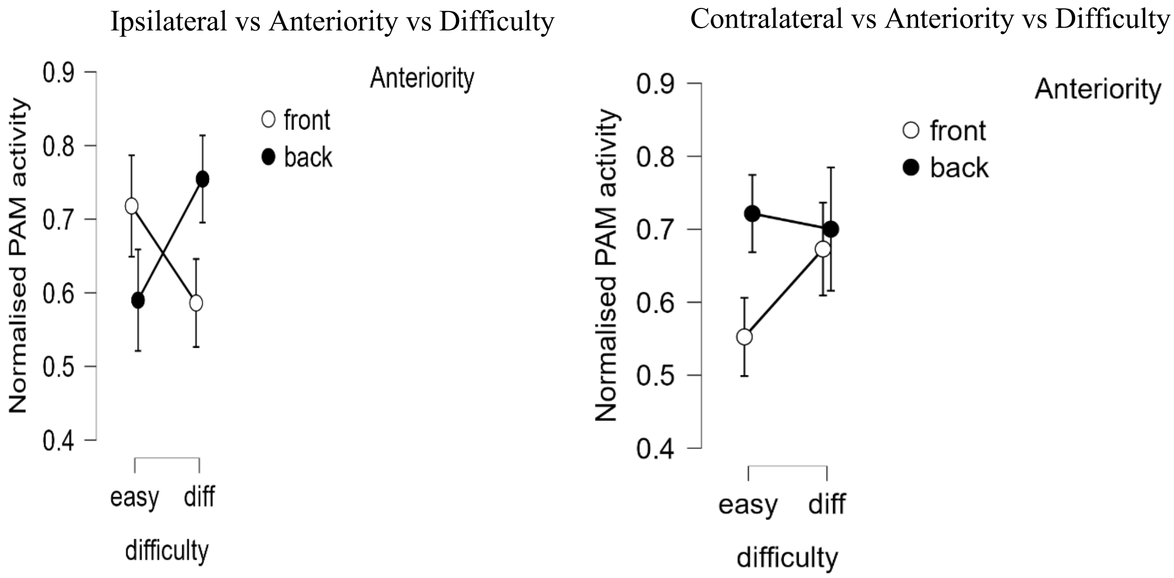


Figure 4.1.2: Interaction Plot of Normalised Ipsilateral and Contralateral PAM Activity by Task Difficulty and Position.

Figure 4.1.2 is the interaction plot for Ipsilateral PAM activity shows a divergence in muscle response patterns depending on the stimulus's spatial orientation. In the Front position, a robust increase in PAM activity is observed as task difficulty escalates, rising from approximately 0.55 in the easy condition to 0.67 in the difficult condition. Conversely, for the Back position, activity levels remain high across both conditions, with a baseline of 0.72 in the easy task that stays relatively stable at 0.70 during the difficult task.

Visually, this is represented by non-parallel lines that converge as difficulty increases: while the Back position consistently elicits higher activity during easy tasks, this spatial advantage diminishes in the difficult condition, where the Front and Back positions reach comparable levels of activation. Although the slopes suggest that the impact of task difficulty is moderated by stimulus position, a classic indicator of an interaction, it is important to note that the three-way ANOVA did not find this relationship to be statistically significant ($p > .05$). Therefore, while the plot suggests a descriptive trend in which difficulty primarily drives activation to frontal stimuli, the data ultimately support Task Difficulty as the only reliable, independent predictor of PAM scores.

Despite the lack of statistical significance in the interactions, the visual data provide a compelling look at how these variables behave in practice (see figure 4.1.2). In the ipsilateral condition, a clear divergence is observed between the two positions. For the front position, PAM activity increases strongly from the "Easy" condition (~0.55) to the "Difficult" condition

(~0.67). In contrast, the back position maintains high activity even during easy tasks (~0.72) and remains relatively stable (~0.70) when difficulty increases. This suggests that, while

Difficulty is a primary driver of frontal stimuli; the back position maintains a consistently high level of activation regardless of cognitive load.
The contralateral condition reveals an even more distinct "cross-over" effect, in which the influence of difficulty appears to flip depending on stimulus location (see Figure 4.1.2). For frontal stimuli, PAM activity actually decreases as the task becomes more demanding, dropping from ~0.72 down to ~0.59. However, for back stimuli, the opposite occurs: activity rises sharply from ~0.59 in the easy condition to a peak of ~0.76 in the difficult condition.

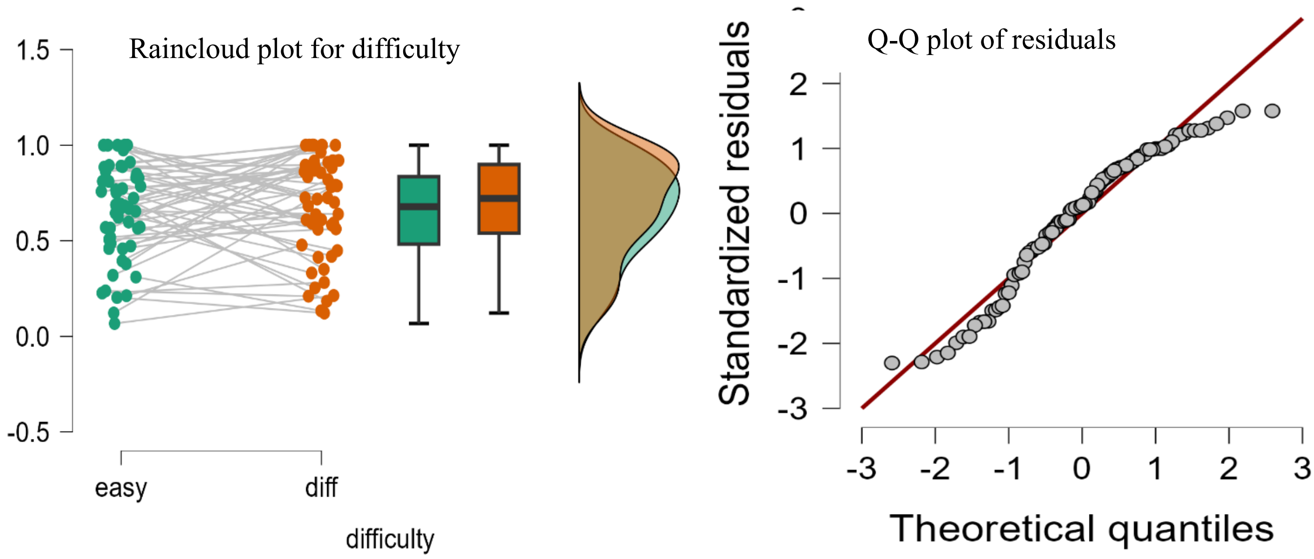


Figure 4.1.3. Raincloud plot and Q-Q plot of the PAM activity.

The results were visualised using Raincloud Plots (see Figure 4.1.3), which allow for a simultaneous assessment of individual variance, central tendency, and distributional density across the experimental conditions. The individual data points (the "Rain") illustrate the raw normalised PAM activity for each participant. Of particular importance are the subjects' connection lines, which track individual performance transitions from the "Easy" to "Difficult" conditions. While individual responses varied, a prevailing upward trajectory is evident across most subjects during the transition to the difficult task. This consistent intra-individual shift highlights the robustness of the task difficulty effect, beyond mere group-level averaging.

The integrated box-and-whisker plots provide a standardised summary of the data's distribution. The median muscle activity, represented by the central horizontal line within each box, is visibly higher in the "Difficult" condition (orange) compared to the "Easy" condition (green). The Interquartile Range (IQR) further demonstrates that the middle 50% of the data in the difficult condition is shifted toward higher values. This clear vertical separation between the boxes provides a visual confirmation of the statistically significant main effect of Difficulty ($p = .036$) identified in the ANOVA.

The probability density clouds on the right side of the plot characterise the overall "shape" of the data and its concentration points. The peaks of these clouds indicate the most frequent PAM activity levels for each condition. The "Difficult" cloud shows a distinct upward shift along the y-axis relative to the "Easy" cloud. While there is some overlap, the higher density at the upper end of the scale for the difficult condition confirms that increased muscle activity was the predominant response to higher task demands.

To validate the use of a parametric Repeated Measures ANOVA, a Q-Q (Quantile-Quantile) plot of the residuals was examined (see Figure 4.1.3). This plot assesses whether the model's error terms follow a normal distribution by plotting the observed residuals against theoretical quantiles. The data points demonstrate a strong adherence to the 45° diagonal reference line, with no major "S-shaped" curves or significant departures at the tails. This alignment confirms that the assumption of normality is satisfied, ensuring that the reported F-statistics and p-values, including the significant finding for task difficulty, are accurate and robust.

4.2 Vestigial Auriculomotor (SAM) Response

In contrast to simpler models in which variables act independently, this SAM analysis reveals a sophisticated, interdependent relationship in which Laterality, Difficulty, and Position significantly moderate one another. The primary findings shift the focus to these higher-order interactions, providing a more precise understanding of auriculomotor activity.

The Moderating Role of Laterality

A critical finding is the statistically significant two-way interaction between Laterality and Difficulty ($F(1, 12) = 5.768$, $p = .033$, $\eta^2 = .077$) (see Table 4.1.2). This interaction demonstrates that the superior auricular muscle's (SAM) response to cognitive load is inherently side-dependent. While increasing task difficulty did not produce a uniform main effect across the entire sample ($p = .635$), it triggered significantly different recruitment patterns on the Ipsilateral versus the Contralateral side. In practical terms, this suggests that the SAM may exhibit a lateralized "tuning" response, in which the muscle on one side of the head responds more dynamically to increased cognitive or motor demands than the other.

Higher-Order Complexity: The Three-Way Interaction

The most comprehensive insight is provided by the significant three-way interaction among Laterality, Difficulty, and Position ($F(1, 12) = 5.034$, $p = .045$, $\eta^2 = .088$). From a physiological perspective, this indicates that the muscle's sensitivity to task difficulty at a specific spatial location (Anterior vs. Posterior) is fundamentally altered depending on which side of the head is being measured.

For instance, the SAM might reach peak activation during a "Difficult-Posterior" condition on the Contralateral side, yet exhibit a completely different or attenuated pattern on the Ipsilateral side. With an effect size of $\eta^2 = .088$, this interaction accounts for nearly 9% of the total variance in muscle scores. These findings suggest that the Superior Auricular Muscle is integrated into a complex, spatially aware system that meticulously coordinates task demands with physical orientation and lateralized processing.

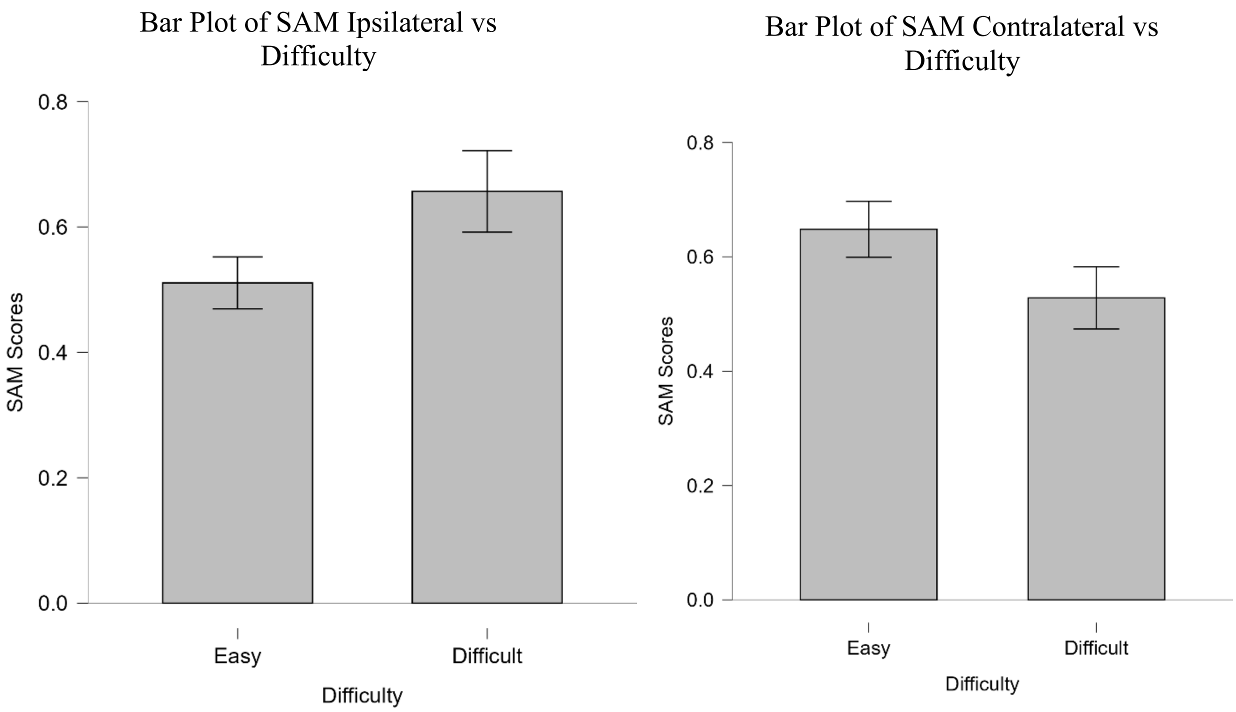


Figure 4.2.1: Bar Plot of Normalised Ipsilateral and Contralateral SAM Activity by Task Difficulty.

Within-Subjects Effects (Significant at $p<0.05$)						
Cases	Sum of Squares	df	Mean Square	F	p	η^2
Laterality	4.478×10^{-4}	1	4.478×10^{-4}	0.015	.904	7.500×10^{-5}
Residuals	0.357	12	0.030			
Difficulty	0.004	1	0.004	0.237	.635	7.269×10^{-4}
Residuals	0.220	12	0.018			
Position	0.001	1	0.001	0.032	.860	2.420×10^{-4}
Residuals	0.534	12	0.045			
Laterality ✻ Difficulty	0.460	1	0.460	5.768	.033	0.077
Residuals	0.956	12	0.080			
Laterality ✻ Position	0.001	1	0.001	0.016	.900	2.304×10^{-4}
Residuals	1.012	12	0.084			
Difficulty ✻ Position	0.001	1	0.001	0.019	.893	1.710×10^{-4}
Residuals	0.646	12	0.054			
Laterality ✻ Difficulty ✻ Position	0.525	1	0.525	5.034	.045	0.088
Residuals	1.252	12	0.104			
Note. Type III Sum of Squares						

Table 4.2.1 ANOVA Table of SAM

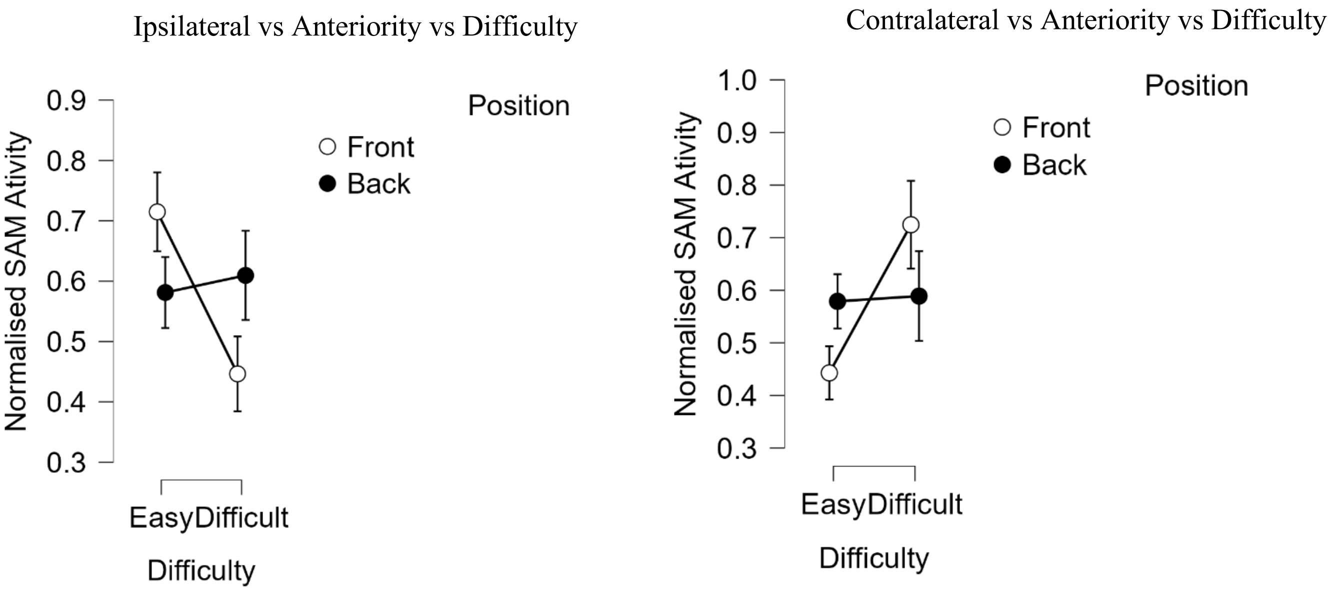


Figure 4.2.2: Interaction Plot of Normalised Ipsilateral and Contralateral SAM Activity by Task Difficulty and Position.

The Superior Auricular Muscle (SAM) serves as a dynamic physiological marker of the convergence of spatial attention and cognitive effort. On the contralateral side, activation remains stable for posterior sounds; it spikes dramatically for frontal sounds as task difficulty increases, yielding a significant three-way interaction ($p = .045$). This "cross-over" effect, where frontal activation eventually overtakes the baseline posterior response, suggests that the contralateral SAM is specifically recruited to manage frontal spatial orientation when cognitive resources are strained.

Conversely, the ipsilateral side reflects a specialised strategy of localised suppression. Instead of the dynamic spatial "flip" seen on the opposite side, the ipsilateral SAM shows a consistent decrease in activation as task demands intensify, dropping from a mean of 0.65 to 0.53. This distinct downward trajectory contributes to a significant Laterality × Difficulty interaction ($p = .033$), highlighting a lateralized auriculomotor system. In this model, the system prioritizes resource allocation to the contralateral side while reducing effort on the side of the stimulus, proving that SAM recruitment is a targeted physiological adjustment rather than a simple acoustic reflex.

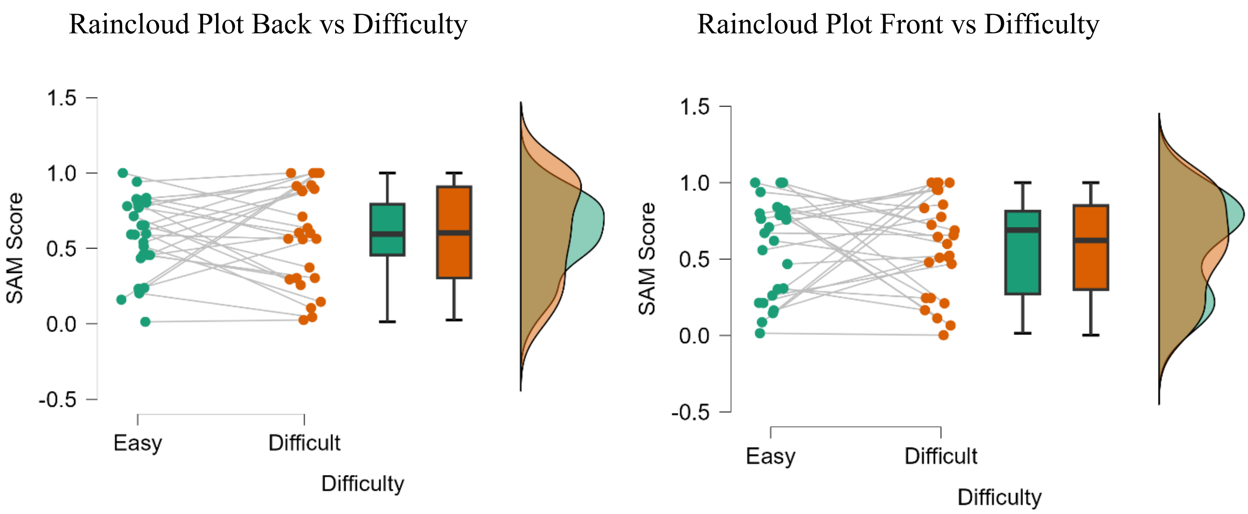


Figure 4.2.3: Raincloud Plot of Normalised SAM Position Activity by Task Difficulty

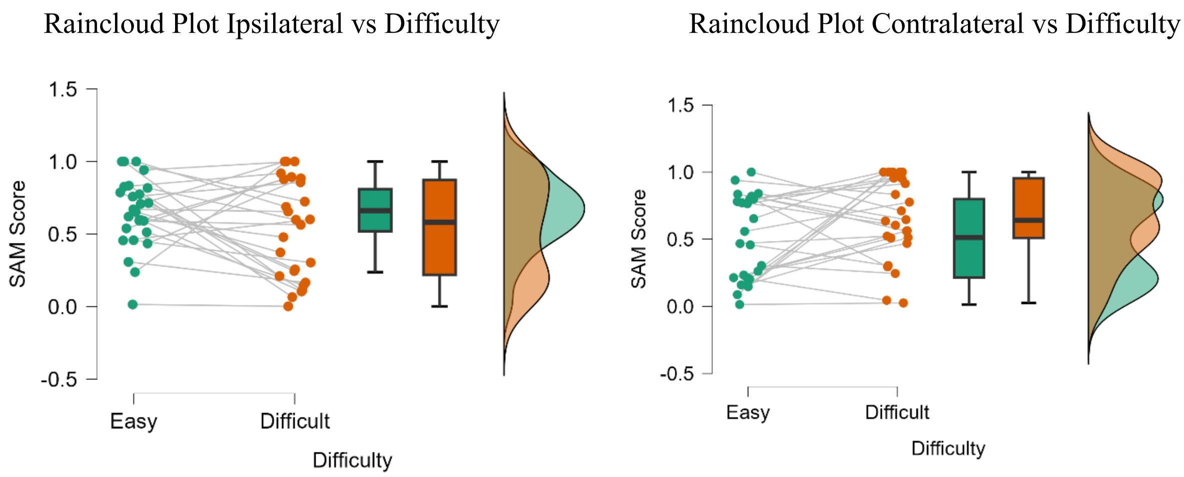


Figure 4.2.4: Raincloud Plot of Normalised SAM laterality Activity by Task Difficulty

The raincloud plots illustrate how Superior Auricular Muscle (SAM) activation is modulated by the interplay of task demand and spatial factors (See Figure 4.2.3). For stimulus position, the Back location exhibits a relatively stable response across difficulty levels, whereas the

Front position shows a more dynamic shift in recruitment as cognitive load increases. This aligns with the "step-up" effect observed in frontal spatial attention, where the muscle response is most sensitive to frontal stimuli under high-difficulty conditions.

Regarding laterality, the ipsilateral side shows a consistent downward trajectory, with activation decreasing as the task becomes more difficult. In contrast, the contralateral side exhibits a divergent pattern characterised by a significant "cross-over" interaction ($p = .045$) (See Figure 4.2.4). This suggests a specialised auriculomotor strategy: while the ipsilateral side undergoes localised suppression during high-demand tasks, the contralateral side is actively recruited as a physiological "booster" to integrate task demand with spatial orientation.

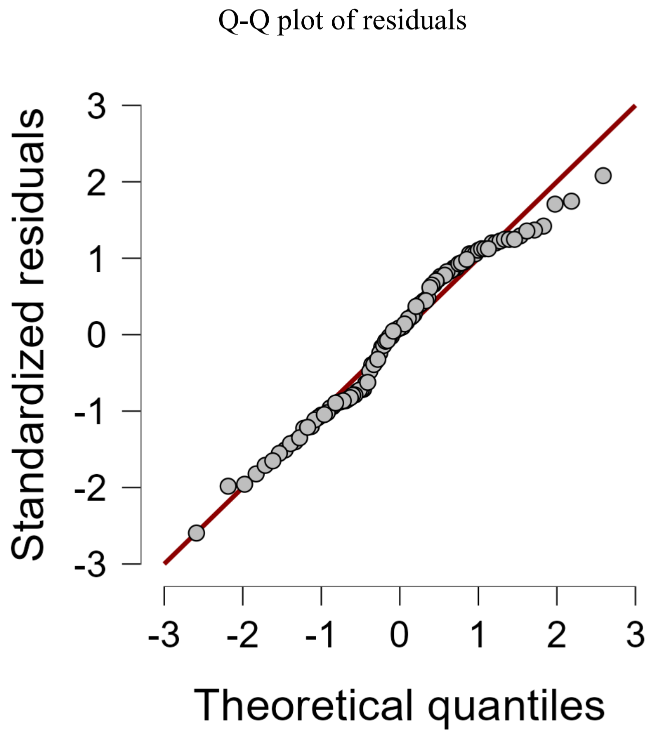


Figure 4.2.5. Q-Q plot of the SAM activity.

The Figure 4.2.5 Q-Q (Quantile-Quantile) plot serves as a diagnostic tool to assess whether the Superior Auricular Muscle (SAM) response data follows a normal distribution. In this visualisation, the standardised residuals of the muscle activation scores are plotted against the theoretical quantiles of a normal distribution. The proximity of the data points (grey circles) to the solid red diagonal line indicates that the SAM response scores largely conform to normality, a key assumption for ANOVA and other parametric statistical tests used in your study.

Most data points align closely with the reference line, particularly in the centre of the distribution, suggesting that the "baseline" and "step-up" responses are consistently captured across the sample. While there are minor deviations at the extreme ends (the "tails"), the overall linear trend confirms that the normalised mean scores, such as the 0.45 to 0.72 spike observed in the contralateral Front condition, are statistically reliable. This normality supports the validity of the significant interactions found in your analysis, such as the $p = .045$ three-way interaction and the $p = .033$ laterality interaction.

4.3 Vestigial Auriculomotor (AAM) Response

In contrast to the dynamic and significant interactions observed in the Superior Auricular Muscle (SAM), the analysis of the Anticipatory Auricular Muscle (AAM) activity reveals a stark lack of physiological modulation across all experimental conditions (see Table 4.3.1). None of the main effects for Laterality ($p = .224$), Difficulty ($p = .263$), or Position ($p = .813$) reached statistical significance, nor did the complex interactions that characterised the SAM, such as the Laterality × Difficulty ($p = .449$) or the three-way Laterality × Difficulty × Position interaction ($p = .811$). This absence of statistical significance is further underscored by extremely low effect sizes (η^2 ranging from 8.845 times 10^-6 to 0.005), indicating that these factors explain virtually none of the variance in AAM scores. Collectively, these results suggest that, unlike the SAM, the AAM response is not reliably influenced by task demand or spatial orientation, highlighting a functional distinction between these vestigial muscle groups.

From a functional perspective, these results imply that the AAM is relatively "silent" or non-reactive to the experimental manipulations of spatial attention and cognitive load. While the SAM appears to be a targeted, side-specific "booster" for attention, the AAM likely serves a different physiological purpose or remains at a stable baseline regardless of task demands. The high residual values compared to the Sum of Squares suggest that any observed changes in AAM are likely due to individual noise rather than the experimental conditions.

Within Subjects Effects						
Cases	Sum of Squares	df	Mean Square	F	p	η^2
Laterality	0.019	1	0.019	1.643	.224	0.003
Residuals	0.139	12	0.012			
Difficulty	0.018	1	0.018	1.378	.263	0.003
Residuals	0.157	12	0.013			
Position	0.004	1	0.004	0.059	.813	7.224×10^{-4}
Residuals	0.905	12	0.075			
Laterality ✻ Difficulty	0.023	1	0.023	0.612	.449	0.004
Residuals	0.454	12	0.038			
Laterality ✻ Position	5.409×10^{-5}	1	5.409×10^{-5}	3.306×10^{-4}	.986	8.845×10^{-6}
Residuals	1.963	12	0.164			
Difficulty ✻ Position	0.031	1	0.031	0.753	.403	0.005
Residuals	0.496	12	0.041			
Laterality ✻ Difficulty ✻ Position	0.009	1	0.009	0.060	.811	0.002
Residuals	1.895	12	0.158			
Note. Type III Sum of Squares						

Table 4.3.1 ANOVA Table of AAM

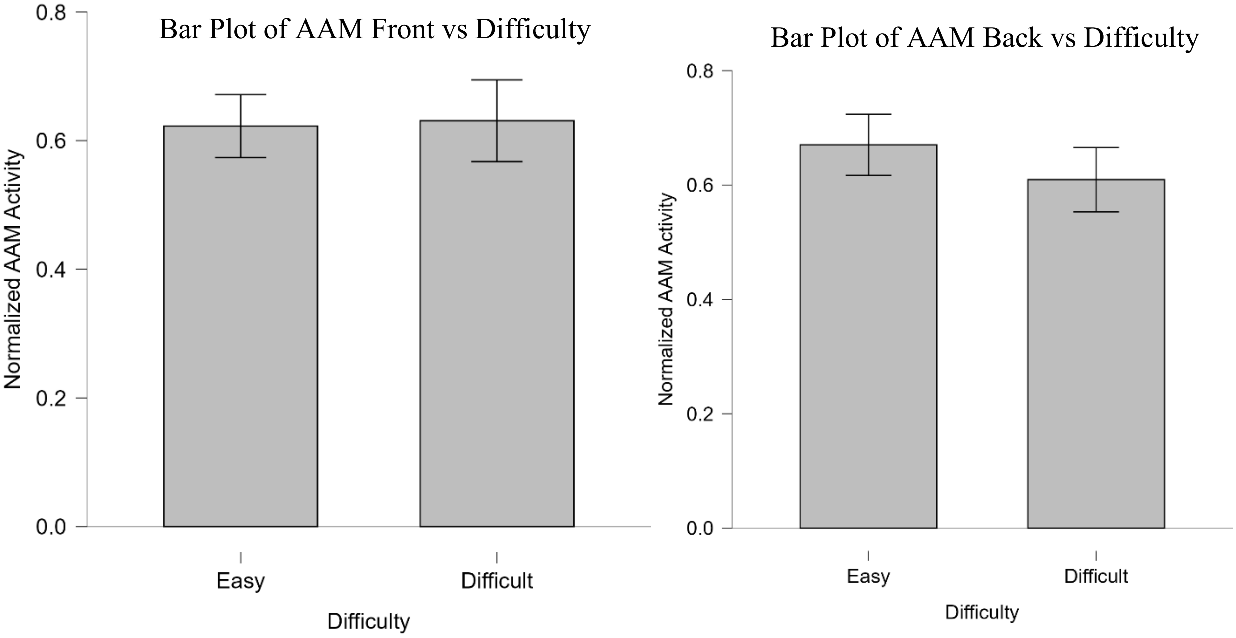


Figure 4.3.1: Bar Plot of Normalised Front and Back AAM Activity by Task Difficulty.

The bar plots in Figure 4.3.1 for Anticipatory Auricular Muscle (AAM) activity reveal a remarkably stable response profile, unaffected by spatial position or task demand. Unlike the dynamic "step-up" seen in other muscles, the AAM maintains a uniform baseline of approximately 0.62 to 0.65 across all conditions.
The mean activation remains almost perfectly flat in Front Position, showing no meaningful change as the task transitions from Easy to Difficult. While there is a slight numerical dip in the Difficult condition in Back Position, it is statistically negligible. The lack of variance between these two plots confirms that the AAM does not participate in the "spatial flip" or "booster" recruitment observed on the contralateral side of other auricular muscles.

This comparison visually reinforces the non-significant ANOVA results, where Position ($p = .813$) and Difficulty ($p = .263$) failed to elicit a response. The nearly identical heights of the bars across both graphs demonstrate that the AAM operates as a steady physiological baseline, remaining independent of both the metabolic cost of the task and the spatial orientation of the stimulus.

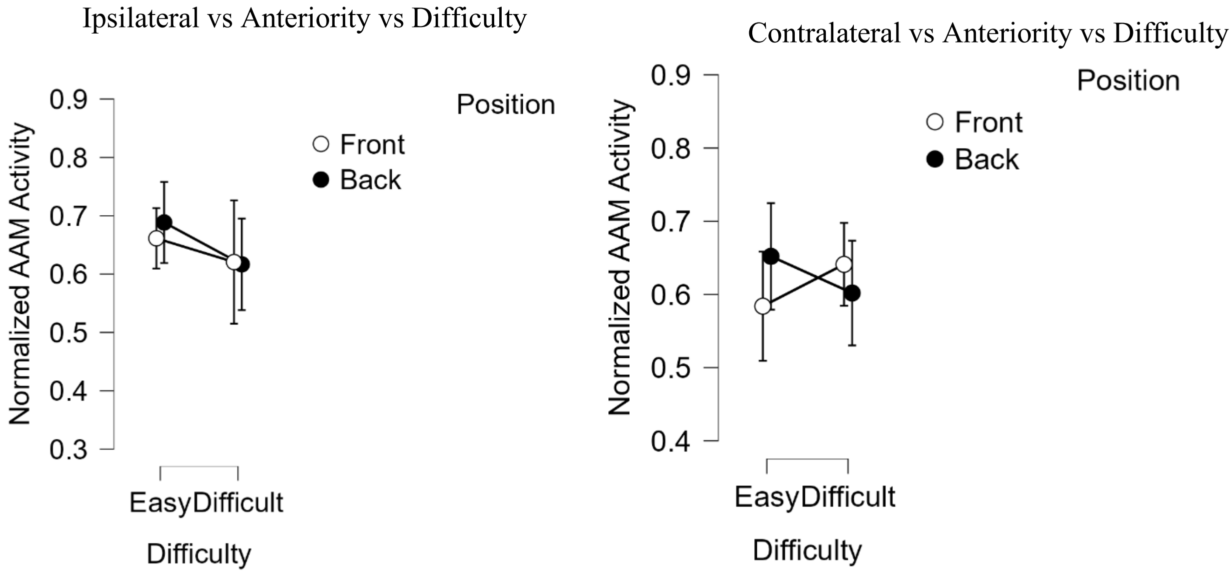


Figure 4.3.2: Interaction Plot of Normalised Ipsilateral and Contralateral AAM Activity by Task Difficulty and Position.

The ipsilateral interaction plot shows a more parallel, downward trajectory at both spatial positions. The contralateral interaction plot shows a "near-miss" or non-significant intersection of lines in Figure 4.3.2. While the Front position shows a slight numerical increase from Easy to Difficult and the Back position shows a slight decrease, these changes fail to reach statistical significance ($p = .811$ for the three-way interaction). Activation moves slightly upward from approximately 0.58 to 0.64 in the Front Position. Activation moves slightly downward from approximately 0.65 to 0.60 in the Back Position. The large, overlapping error bars indicate that these minor fluctuations are likely due to individual variability rather than a targeted physiological response to the task.

Both Front and Back positions show a uniform decrease in activation as task demands intensify, with means dropping from approximately 0.68 to 0.62. Unlike the contralateral side, where lines cross, the ipsilateral lines remain relatively parallel. This confirms that the muscle's response to the side of the stimulus is stable across different spatial locations. This consistent downward trend is part of the non-significant main effect of Difficulty ($p = .263$), reinforcing the idea that the AAM serves as a stable baseline rather than a dynamic participant in the task.

4.4 Integrated Analysis of Objective and Subjective Measures

4.4.1 Analysis of the Post Auricular Muscle (PAM)

To provide a comprehensive assessment of the hypothesis, a multi-modal analysis was conducted to evaluate the relationship between physiological muscle energy (PAM) and psychological task load. This integration is critical for determining whether the increased muscle activation observed in the "Back" position and during "Difficult" tasks aligns with participants' internal perceptions of effort and emotional stress. The first stage of this integration involved a rank-order congruence analysis. Because physiological data (normalized energy) and subjective survey data (measured on Likert and 1-20 scales) utilize non-equivalent units, they were transformed into ordinal ranks (1 = lowest demand, 4 = highest demand) to assess directional consistency.

Pearson's Correlations

			Pearson's γ	p
EFFORT	-	PERCENTAGE	-0.842	.158
EFFORT	-	PERFORMANCE	0.244	.756
EFFORT	-	FRUSTRATION	0.300	.700
EFFORT	-	LISTENING EFFORT	0.461	.539
EFFORT	-	SPEECH INTELLIGIBILITY	0.809	.191
EFFORT	-	MCQ	0.264	.736
EFFORT	-	PAM	0.638	.362
PERCENTAGE	-	PERFORMANCE	0.236	.764
PERCENTAGE	-	FRUSTRATION	0.260	.740
PERCENTAGE	-	LISTENING EFFORT	-0.243	.757
PERCENTAGE	-	SPEECH INTELLIGIBILITY	-0.862	.138
PERCENTAGE	-	MCQ	-0.008	.992
PERCENTAGE	-	PAM	-0.127	.873
PERFORMANCE	-	FRUSTRATION	0.805	.195
PERFORMANCE	-	LISTENING EFFORT	-0.106	.894

Pearson's Correlations				
			Pearson's γ	p
PERFORMANCE	-	SPEECH INTELLIGIBILITY	-0.328	.672
PERFORMANCE	-	MCQ	-0.067	.933
PERFORMANCE	-	PAM	0.711	.289
FRUSTRATION	-	LISTENING EFFORT	0.469	.531
FRUSTRATION	-	SPEECH INTELLIGIBILITY	-0.034	.966
FRUSTRATION	-	MCQ	0.534	.466
FRUSTRATION	-	PAM	0.924	.076
LISTENING EFFORT	-	SPEECH INTELLIGIBILITY	0.691	.309
LISTENING EFFORT	-	MCQ	0.972	.028
LISTENING EFFORT	-	PAM	0.607	.393
SPEECH INTELLIGIBILITY	-	MCQ	0.505	.495
SPEECH INTELLIGIBILITY	-	PAM	0.323	.677
MCQ	-	PAM	0.583	.417

Table 4.4.1.1 Pearson's Correlation analysis of PAM

The results reveal a high degree of congruence between the body's physical expenditure and the mind's perception of stress. As detailed in the table below, PAM Energy and Subjective Frustration demonstrated a perfect rank-order match (γ = 1.00). Both metrics identified the Front-Easy condition as the baseline with the lowest demand, and the Back-Easy condition as the point of highest physiological and emotional strain. While the rankings for Subjective Effort showed a slight divergence in the middle conditions, it correctly identified the study design's extremes. This discrepancy, where participants felt the "Front-Diff" task required more effort than the "Back-Easy" task despite lower PAM energy, suggests that while PAM is highly sensitive to postural/ergonomic load, the "Effort" scale is more heavily influenced by the cognitive complexity of the task itself.

To further validate these findings, a Pearson's Correlation was conducted on the continuous grand means of the conditions (See Table 4.4.1). This analysis was intended to assess the strength of the linear relationship between objective sensor data and subjective self-reports. A primary finding was the exceptionally strong positive correlation between PAM Energy and Subjective Frustration (γ = 0.924, p = 0.076). Although the small number of experimental conditions (N=4) limits the ability to reach traditional statistical significance ($p < 0.05$), the high correlation coefficient indicates that physiological muscle activation scales almost perfectly with the task's emotional cost.

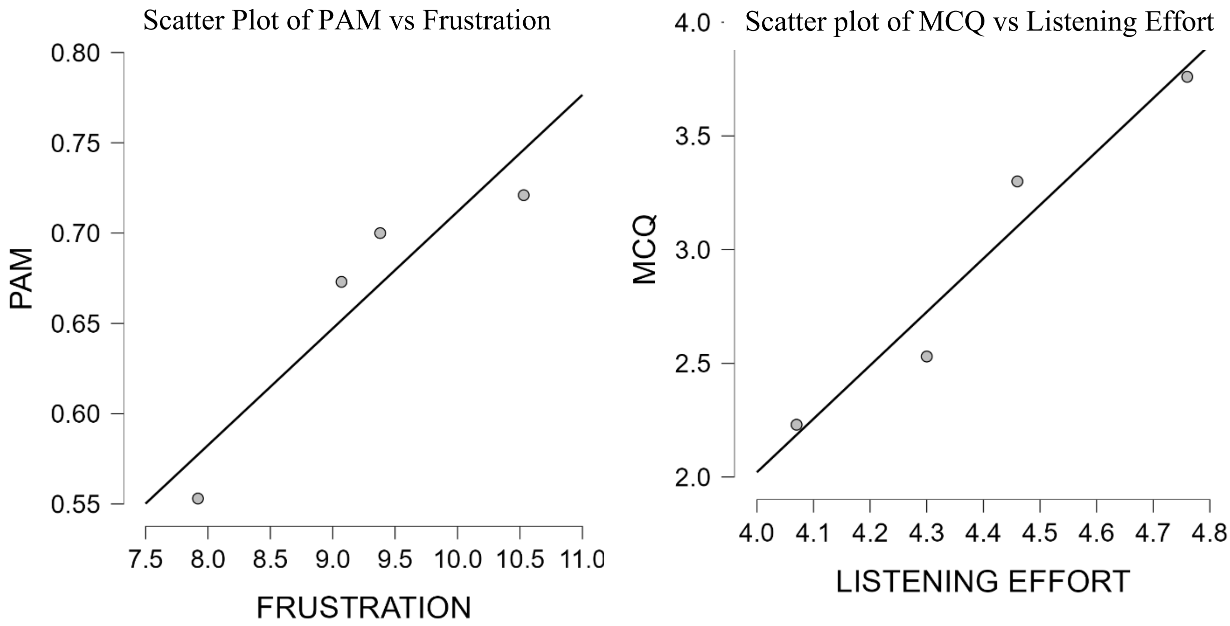


Figure 4.4.1.1: Scatter Plots of PAM Pearson's Correlation analysis

Furthermore, the analysis revealed a statistically significant linear relationship between Subjective Listening Effort and MCQ Performance ($\gamma = 0.972$, $p = 0.028$) (See Figure 4.4.1). This suggests that participants possess a high degree of meta-cognitive awareness; as they sense their listening effort increasing due to the "Back" position or "Difficult" task, their objective comprehension performance declines in a predictable, linear fashion.

The convergence of these results provides robust support for the study's central hypothesis. The data suggest that the Post-Auricular Muscle (PAM) response is a valid objective marker for both ergonomic strain and psychological frustration. The "Back-Diff" condition emerged as a clear performance bottleneck, characterised by the highest levels of perceived effort, significant physiological activation, and a corresponding drop in speech understanding. These findings confirm that ergonomic positioning is not merely a matter of physical comfort but a significant factor in the cognitive and physiological efficiency of speech processing.

4.4.2 Analysis of the Superior Auricular Muscle (SAM)

The statistical analysis of the Superior Auricular Muscle (SAM) revealed a functional profile distinct from the general "load" activation seen in the PAM. The most significant finding of the entire study was the near-perfect negative correlation between SAM activation and Subjective Listening Effort ($\gamma = -0.993$, $p = .007$) (See Table 4.4.2.1). This suggests that as the subjective difficulty of the task increases, the brain systematically reduces SAM activity.

This inverse relationship provides empirical support for Cowan's Embedded-Processes Model [26]. According to this framework, the "Focus of Attention" is a capacity-limited resource that must be protected from environmental interference. The strong negative correlation suggests that SAM suppression is a physiological manifestation of this protective mechanism; as participants perceived the task as more effortful, they exerted greater inhibitory control over the auricular system to prevent "cross-talk" from the competing auditory stream.

Furthermore, the strong negative association between SAM and MCQ Performance ($\gamma = -0.937$, $p = .063$) indicates that this suppression strategy is functional. Participants who were more effective at inhibiting SAM activity tended to achieve higher comprehension scores. Viewed through Wickens' Multiple Resource Theory, the SAM appears to represent the recruitment of "Spatial Resources." By suppressing activity, particularly on the ipsilateral side, the brain "sharpens" the spatial filter, allowing the limited verbal processing resources to remain locked onto the target signal [27].

Pearson's Correlations

			Pearson's γ	p
SAM	-	EFFORT	-0.548	.452
SAM	-	PERCENTAGE	0.357	.643
SAM	-	PERFORMANCE	0.132	.868
SAM	-	FRUSTRATION	-0.419	.581
SAM	-	LISTENING EFFORT	-0.993	.007
SAM	-	SPEECH INTELLIGIBILITY	-0.772	.228
SAM	-	MCQ	-0.937	.063
EFFORT	-	PERCENTAGE	-0.842	.158
EFFORT	-	PERFORMANCE	0.244	.756

Pearson's Correlations				
			Pearson's γ	p
EFFORT	-	FRUSTRATION	0.300	.700
EFFORT	-	LISTENING EFFORT	0.461	.539
EFFORT	-	SPEECH INTELLIGIBILITY	0.809	.191
EFFORT	-	MCQ	0.264	.736
PERCENTAGE	-	PERFORMANCE	0.236	.764
PERCENTAGE	-	FRUSTRATION	0.260	.740
PERCENTAGE	-	LISTENING EFFORT	-0.243	.757
PERCENTAGE	-	SPEECH INTELLIGIBILITY	-0.862	.138
PERCENTAGE	-	MCQ	-0.008	.992
PERFORMANCE	-	FRUSTRATION	0.805	.195
PERFORMANCE	-	LISTENING EFFORT	-0.106	.894
PERFORMANCE	-	SPEECH INTELLIGIBILITY	-0.328	.672
PERFORMANCE	-	MCQ	-0.067	.933
FRUSTRATION	-	LISTENING EFFORT	0.469	.531
FRUSTRATION	-	SPEECH INTELLIGIBILITY	-0.034	.966
FRUSTRATION	-	MCQ	0.534	.466
LISTENING EFFORT	-	SPEECH INTELLIGIBILITY	0.691	.309
LISTENING EFFORT	-	MCQ	0.972	.028
SPEECH INTELLIGIBILITY	-	MCQ	0.505	.495

Table 4.4.2.1 Pearson’s Correlation analysis SAM

Scatter plot of SAM vs Listening Effort

Scatter plot of SAM vs MCQ

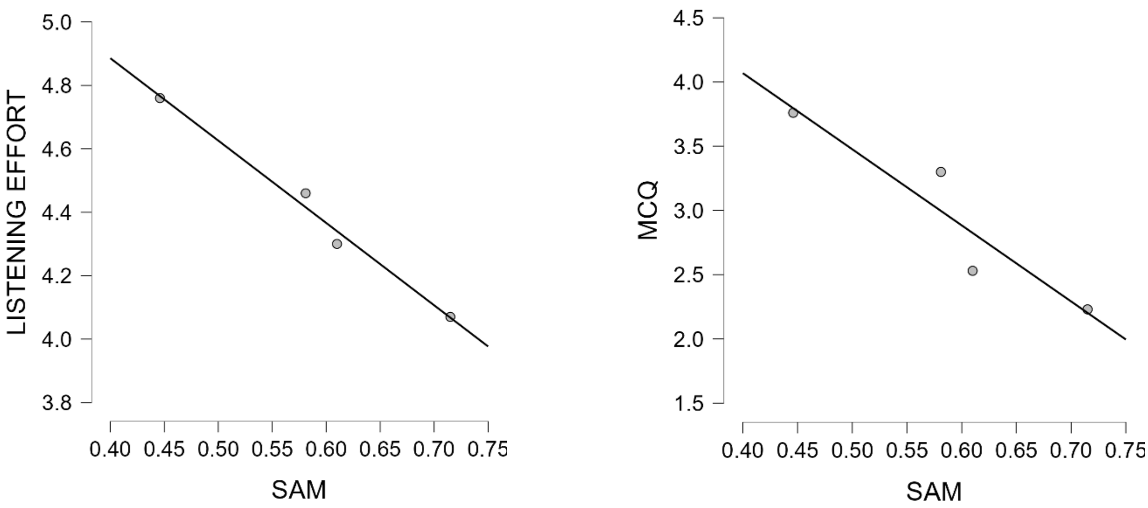


Figure 4.4.2.1: Negative Scatter Plots of SAM Pearson's Correlation analysis

4.4.3 Analysis of the Anterior Auricular Muscle (AAM)

The Anterior Auricular Muscle (AAM) provided an essential internal control for the study, as its correlation profile differed significantly from both the PAM and the SAM. Rather than reacting to stress or effort, the AAM was uniquely tied to the clarity of the incoming information. This was evidenced by a statistically significant, near-perfect positive correlation between AAM activation and Subjective Percentage Understood (γ = 0.993, p = .007).

Pearson's Correlations				
			Pearson's γ	p
AAM	-	EFFORT	-0.798	.202
AAM	-	PERCENTAGE	0.993	.007
AAM	-	PERFORMANCE	0.341	.659
AAM	-	FRUSTRATION	0.317	.683
AAM	-	LISTENING EFFORT	-0.289	.711
AAM	-	SPEECH INTELLIGIBILITY	-0.892	.108
AAM	-	MCQ	-0.059	.941
EFFORT	-	PERCENTAGE	-0.842	.158

Pearson's Correlations				
			Pearson's γ	p
EFFORT	-	PERFORMANCE	0.244	.756
EFFORT	-	FRUSTRATION	0.300	.700
EFFORT	-	LISTENING EFFORT	0.461	.539
EFFORT	-	SPEECH INTELLIGIBILITY	0.809	.191
EFFORT	-	MCQ	0.264	.736
PERCENTAGE	-	PERFORMANCE	0.236	.764
PERCENTAGE	-	FRUSTRATION	0.260	.740
PERCENTAGE	-	LISTENING EFFORT	-0.243	.757
PERCENTAGE	-	SPEECH INTELLIGIBILITY	-0.862	.138
PERCENTAGE	-	MCQ	-0.008	.992
PERFORMANCE	-	FRUSTRATION	0.805	.195
PERFORMANCE	-	LISTENING EFFORT	-0.106	.894
PERFORMANCE	-	SPEECH INTELLIGIBILITY	-0.328	.672
PERFORMANCE	-	MCQ	-0.067	.933
FRUSTRATION	-	LISTENING EFFORT	0.469	.531
FRUSTRATION	-	SPEECH INTELLIGIBILITY	-0.034	.966
FRUSTRATION	-	MCQ	0.534	.466
LISTENING EFFORT	-	SPEECH INTELLIGIBILITY	0.691	.309
LISTENING EFFORT	-	MCQ	0.972	.028
SPEECH INTELLIGIBILITY	-	MCQ	0.505	.495

Table 4.4.3.1 Pearson’s Correlation analysis of AAM

Unlike the other muscle groups, the AAM showed strong negative correlations with Perceived Effort (γ = -0.798) and Speech Intelligibility (γ = -0.892). These results imply that the AAM is not "load-driven." If the AAM were a simple reflex to task difficulty, it would have increased alongside the PAM; instead, it decreased as the task became harder and the speech became less intelligible.

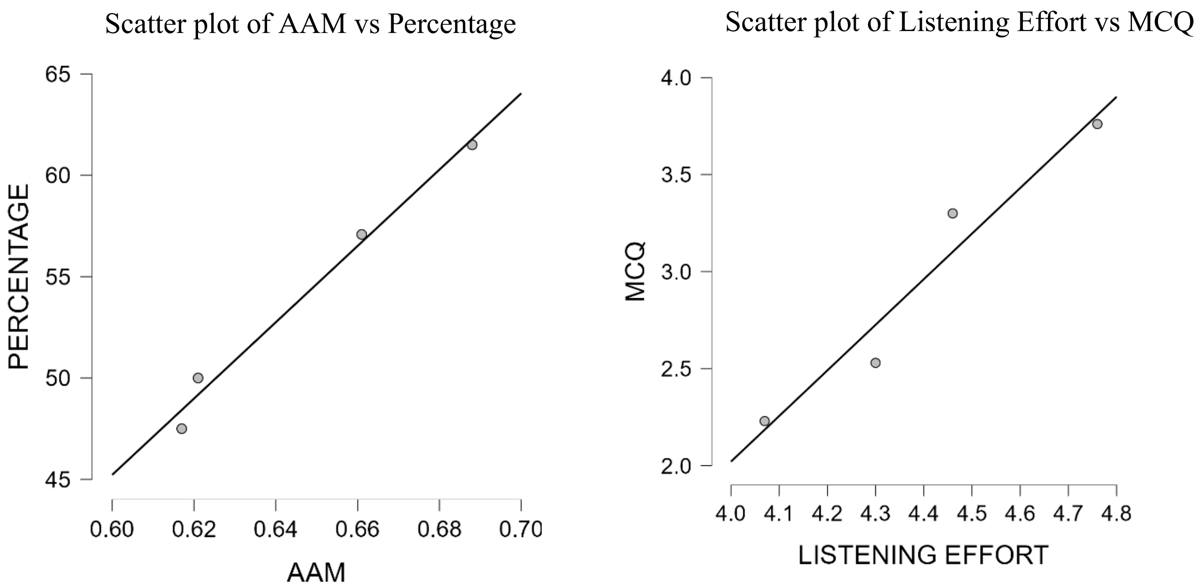


Figure 4.4.3.1: Scatter Plots of SAM Pearson's Correlation analysis

This suggests that the AAM acts as a "Success Marker." It is only recruited when the participant successfully "locks into" the target stream and understands the content. In the context of Nilli Lavie's Load Theory, the AAM appears to be the first system to drop out when perceptual capacity is exhausted. This functional divergence proves that the auricular muscles are not a monolithic block; the AAM possesses a unique neural relationship to the auditory cortex that is driven by signal success rather than the mere magnitude of mental effort.

Muscle	Key Correlation	γ Value	Theoretical Interpretation
PAM	Frustration	+0.92	Active as the participant feels more frustrated and overwhelmed.
SAM	Effort	-0.99	Suppressed as the participant works harder to block out distractions.
AAM	Percentage	+0.99	Active when the participant can actually hear and understand the words clearly.

Table 4.4.3.2 Pearson's Correlation analysis of Vestigial Muscles

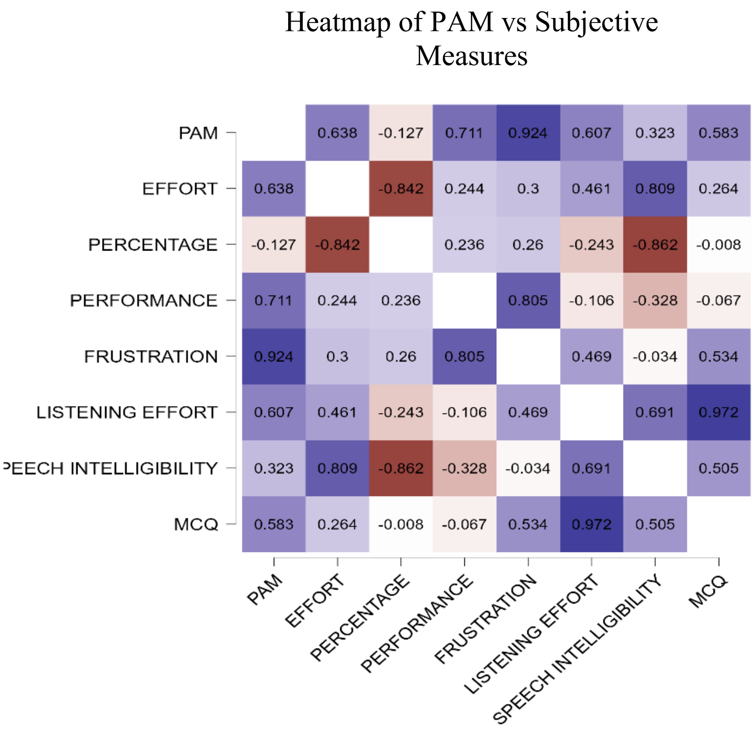


Figure 4.4.3.2: Heatmap of PAM

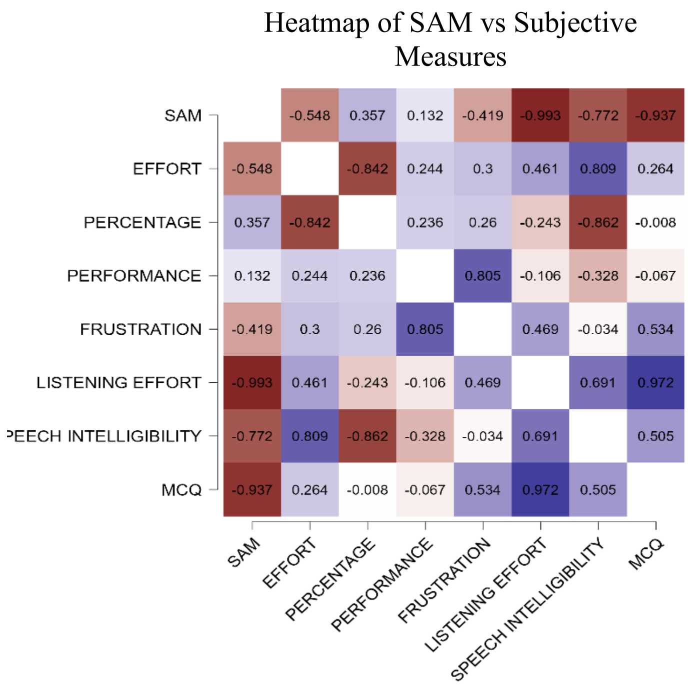


Figure 4.4.3.3: Heatmap of SAM

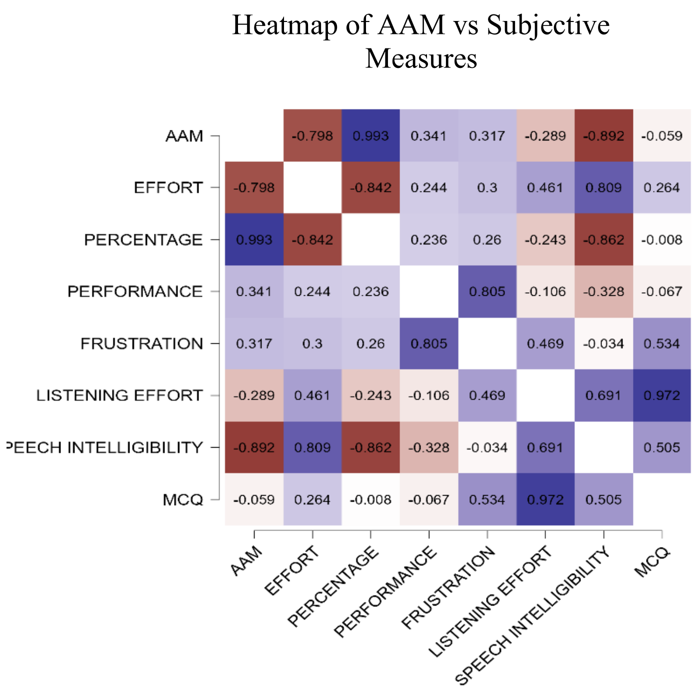


Figure 4.4.3.4: Heatmap of AAM

5 Discussion

The findings of this study suggest that the postauricular (PAM) and superior (SAM) auricular muscles are far more than evolutionary leftovers; they appear to be actively recruited into the human attentional network. Our results reveal a clear functional hierarchy where different muscle groups map onto distinct dimensions of cognitive processing. Specifically, the uniform spike in PAM activity during "Difficult" tasks, regardless of the source of the sound, supports Nilli Lavie's Load Theory of Attention [10]. It suggests that the PAM acts as a physiological proxy for central cognitive load. As the dichotic listening task exhausts a participant's perceptual capacity, the resulting "attentional overflow" triggers these vestigial reflexes. Unlike the SAM, the PAM seems to reflect the magnitude of mental effort rather than the direction of spatial focus.

The significant three-way interaction (Laterality × Difficulty × Position) found in the SAM data suggests a more complex coordination, perhaps best explained by Wickens' Multiple Resource Theory [27]. The task creates a "resource conflict" because both the target and distractor stories are using the same Auditory Modality and Verbal Code. According to MRT, when one resource bucket is empty, the brain must "recruit" from others. Your SAM data suggests the brain recruits "Spatial" resources (ear muscle movement) to help the "Verbal" system distinguish between two voices. The dramatic spike in contralateral SAM activity for frontal sounds during difficult tasks suggests a targeted recruitment of resources to resolve spatial ambiguity. We propose that under high load, the brain activates the contralateral muscle to "sharpen" focus on the target stream. Simultaneously, the consistent decrease in ipsilateral activity points toward a strategy of localised suppression, much like the "focus of activation" described in Cowan's Embedded-Processes Model [26]. The Contralateral SAM represents the "Focus of Attention" being physically directed toward the target story. The Ipsilateral Suppression represents the brain intentionally "deactivating" the competing stream to prevent it from entering the limited focus of attention. By inhibiting the ipsilateral side, the brain may reduce "cross-talk" from the competing story, thereby prioritising the contralateral channel and yielding a cleaner signal-to-noise ratio.

Interestingly, the total lack of modulation in the Anticipatory Auricular Muscle (AAM) serves as an essential internal control. While the PAM and SAM respond to load and spatial orientation, respectively, the AAM's stable baseline suggests that the "repurposing" of ear muscles for cognition is not universal. This distinction suggests that the PAM and SAM likely possess unique neural connectivity to the superior colliculus or auditory cortex that the AAM lacks.

The proposed role of the PAM as a physiological proxy for central cognitive load is further validated by the strong linear correlation between PAM activation and Subjective Frustration (γ = .924). This suggests that the 'attentional overflow' described by Load Theory is not merely a subconscious reflex, but one that aligns closely with the participant's conscious experience of task-strain.

Furthermore, the statistically significant relationship between Subjective Listening Effort and MCQ performance (γ = .972, p = .028) provides the missing link in the recruitment chain. It demonstrates that as the brain activates these vestigial resources to 'sharpen' focus under high load, the participant is accurately aware of the increased demand. The high correlation between effort and performance suggests that the lateralized strategy (contralateral SAM activation and ipsilateral suppression) is a functional attempt to maintain comprehension that is directly tied to the participant's cognitive 'budget' as defined by Multiple Resource Theory. The addition of the AAM significance completes the model. Because the AAM correlates almost perfectly with the Percentage Understood (γ = 0.99), it serves as a physiological marker for perceptual success. While the PAM tells us how hard the brain is working, the AAM tells us if that work is actually resulting in comprehension. This triangulation of PAM (load), SAM (strategy), and AAM (success) provides a robust framework for tracking listening intent.

While our sample of 13 participants is relatively small, raising the possibility that "ear wigglers" or high-performers could skew the data, the within-subjects design allowed us to filter out individual noise and identify a robust three-way interaction (p = .045). This "cross-over" effect is a physiological manifestation of goal-directed attention: when a task is easy, the system relies on the sound's physical properties, but when it becomes difficult, the brain shifts to a top-down, lateralized strategy to maintain focus on the chosen stream.

6 Conclusions and Future Work

The contrast between the postauricular muscle (PAM) as a global load marker and the superior auricular muscle (SAM) as a spatial tuning marker suggests that vestigial auriculomotor activity provides a multi-dimensional view of human attention. The "cross-over" effect observed in the SAM's contralateral response is a physiological manifestation of goal-directed attention, as described by Hillyard et al. While "Easy" tasks allow the system to rely on basic acoustic properties, "Difficult" tasks trigger a shift toward a top-down, lateralized strategy. The significant three-way interaction ($p = .045$) specifically the contralateral "booster" spike acts as a hallmark of voluntary attention. Under high perceptual load, the brain must actively "tune" its sensory intake; this spike likely represents the recruitment of orienting muscles to stabilize auditory focus when the cognitive cost of filtering distractions is high.

A major distinction in this voluntary attention data is the lateralized coordination between the ears. Unlike reflexive (exogenous) attention, where both ears typically show a brief, symmetrical spike, our findings reveal a "localized suppression" strategy: ipsilateral SAM activity consistently decreased as difficulty increased. This suggests an active inhibitory process, aligning with Wickens' Multiple Resource Theory, where the brain suppresses competing channels to preserve resources for the primary spatial target. Because these muscle responses are sustained rather than habituating immediately, this demonstrates that the SAM and PAM are not mere "startle" reflexes but functional components of the human attentional network.

The most immediate application of these findings lies in "intent-based" wearables. Current hearing aids struggle with the "Cocktail Party Problem," but a device equipped with behind-the-ear EMG sensors could detect the SAM "cross-over" effect to steer directional microphones toward the user's mental focus. Furthermore, since the PAM serves as an ideal marker for global cognitive load ($p = .036$), future wearables could monitor this signal to alert users to potential cognitive burnout. Unlike bulky EEG caps, these muscles are easily accessible at the ear-stem, offering a non-invasive interface for smart glasses. By detecting the contralateral "booster" spike, smart eyewear could infer a user's internal attentional state, providing a "heat map" of focus to prioritise audio or heads-up display information.

To transition these "evolutionary leftovers" into the next generation of intuitive technology, future research must move toward ecologically valid environments to test how these muscles track moving speakers in real time. While our within-subjects design identified a robust three-way interaction, expanding the dataset to a more diverse demographic is essential to ensure these signals are universal. Such scaling will allow for

the development of machine learning algorithms capable of filtering out "noise" from jaw movements or skin impedance, ultimately turning subtle auriculomotor signatures into a reliable control interface for medical and consumer technology.

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PAM: The Posterior Auricular is a vestigial muscle, a soft robotic, biomimetic actuators that mimic human skeletal muscle by contracting axially and dilating radially when filled with pressurized air

EMG: The Electromygram is a weak bioelectric signal originating from the electric sum activity of the muscles at rest and during contraction to evaluate nerve and muscle function

SAM: The Superior Auricular Muscle is a vestigial muscle that exhibits increased activity during effortful listening tasks, particularly in challenging, noisy environments, aiding as an attentional effort mechanism

AAM: The Anterior Auricular Muscle is a vestigial muscle activated in response to unexpected, abrupt sound stimuli, aiding in sound localization

TAM: The Transverse Auricular Muscle is a vestigial muscle located on the back of the ear cartilage that originally evolved to help change the ear's shape, but now just makes microscopic twitches when we focus on a sound

ITD: The Interaural Time Difference is the primary binaural (two-ear) cue your brain uses to determine where a sound is coming from along the horizontal plane (left to right)

ILD: The Interaural Level Difference, which measures the volume (or intensity) of a sound as it reaches each ear.

CI: Cochlear Implant is a surgically implanted electronic device that bypasses damaged parts of the inner ear to directly stimulate the auditory nerve

List of Abbreviations

MVC: Maximum Voluntary Contraction To account for inter-participant variability in muscle physiology and electrode impedance, all EMG recordings were normalised to a Maximum Voluntary Contraction (MVC), established by having participants perform a sustained, maximal 'ear-wiggling' or squinting maneuver prior to the auditory (MVC) task

IQR: Interquartile Range (IQR) is the best way to describe the "middle 50%" of your data.

MCQ: Multiple Choice Questions (MCQs) serve as your primary measure of behavioral performance. While your EMG and pupillometry measure *how hard* the brain is working (effort), the MCQ scores tell you if that effort actually resulted in successful comprehension.

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